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Industrial Hygiene Monitoring

1.0 OVERVIEW OF IH MONITORING

Industrial hygiene monitoring in the semiconductor industry is a very broad subject that covers both chemical and physical hazards. It is complicated by the fact that semiconductor operators and technicians rarely stay at one work station for the entire day. Figure 2.1 identifies the flow of a wafer through one of the major process steps (lithographic process).^[1] Figure 2.2 shows the critical lithographic process in relationship to the overall process flow for a typical integrated circuit (IC).^[2] As can be seen from the complexity of the processing steps represented by these figures, wafer movement requires employees to work in *flow-lines*, moving physically from one work station to another. Even when they are stationary for the whole shift, unless a personal sample is taken in their “breathing zone,” there can be significant concentration differences between the sample location and what the person is breathing. Chapter 4 (Ventilation) will provide additional information on air flow characteristics in a typical fab area that significantly affect airborne exposures.

Appendices A–D provide a further review of the semiconductor integrated circuit production process, and should be consulted for an overview of the process chemistries that may be encountered.

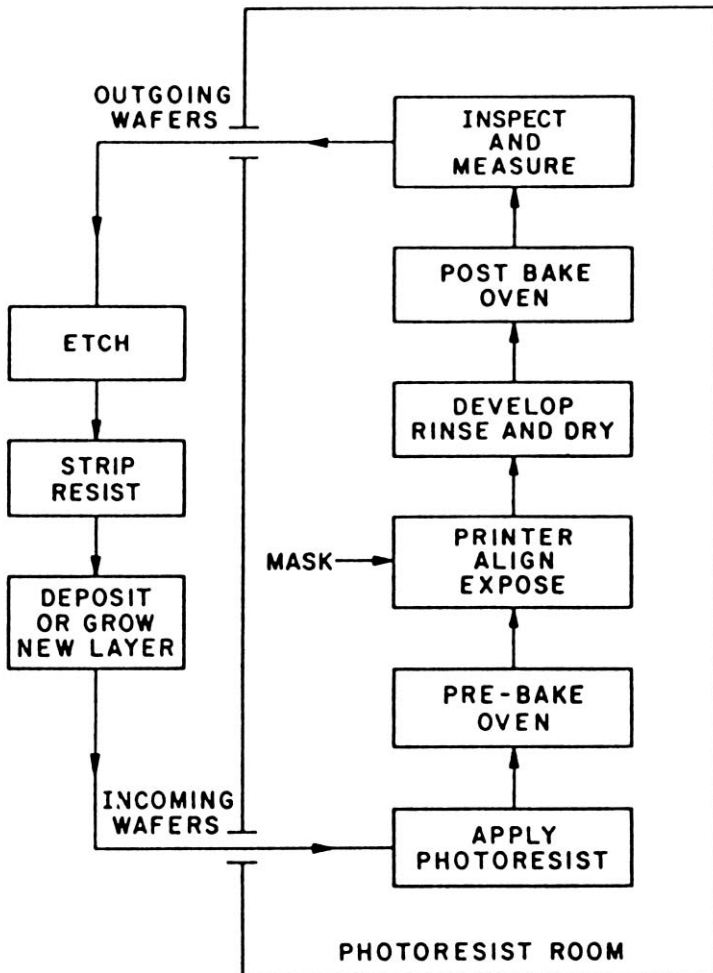


Figure 2.1. Flow of silicon wafer through the lithographic IC processing area.^[1] (McGillis, D. A., Lithography in VLSI Technology, ©1988, McGraw-Hill, Inc. Reprinted with permission.)

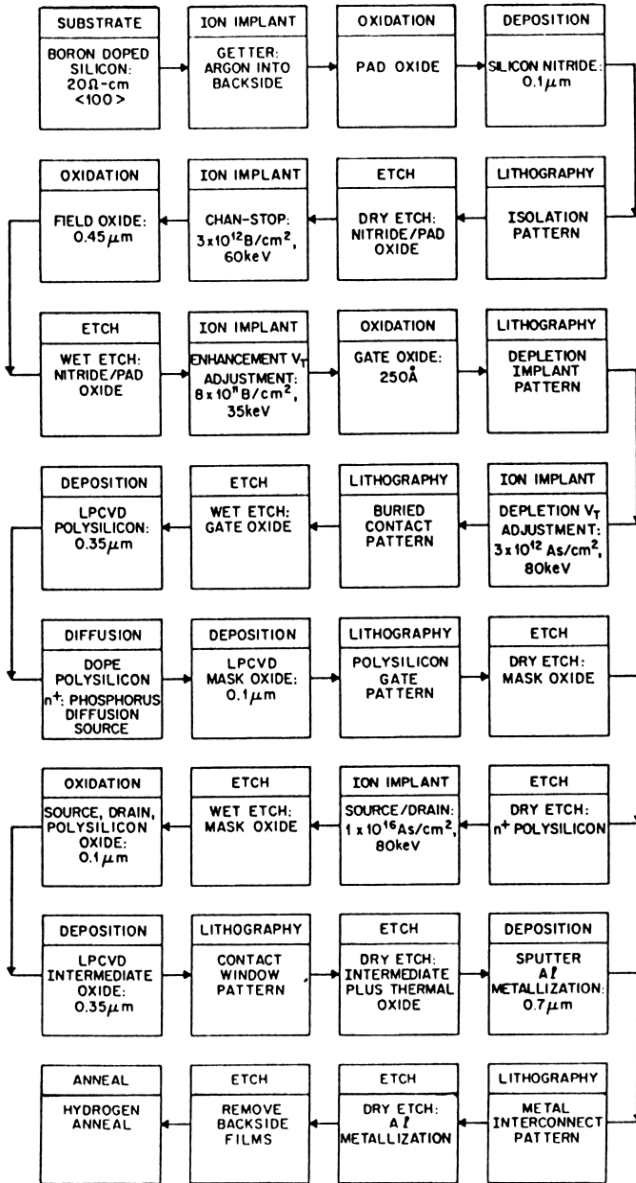


Figure 2.2. Main process steps in an MOS IC process flow.^[2] (Parrillo, L. C., VLSI Process Integration in VLSI Technology, ©1983, McGraw-Hill, Inc. Reprinted with permission.)

1.1 When to Monitor

Normally there are four types of non-emergency events which initiate the need to perform monitoring:

- Complaints (e.g., odors, symptoms without odors, general concern after reading article or media story, etc.)
- New processes or changes in processes
- Routine monitoring
- Odors or other “transient” smells

The complex chemistries and large quantities of hazardous materials that are integral to semiconductor manufacturing increase the level of chemical concerns among cleanroom employees. Quick response to these concerns is often critical. Quick answers may ease concerns or they may trigger rapid action to resolve a hazardous exposure. Also, they help to minimize costly production downtime caused by evacuations.

Before a chemical arrives at the delivery dock, the site needs to know that the material is coming to ensure basic controls are in place to safely use the chemical and to determine if there will be a need to monitor employee exposure. Most semiconductor manufacturers do this as part of a health and safety review and approval procedure for a new process coming on-site. Besides identifying unacceptable exposures, evaluating the process before it goes into production minimizes the need for costly retrofits after the equipment is integrated into the manufacturing process. Chapter 9 (Plan Review) provides a more complete discussion of the review process, including the decommissioning of equipment after it has come in contact with process chemicals.

For this initial evaluation, the chemical purchaser must have a Material Safety Data Sheet (MSDS) on the chemical before the purchase order is processed. Some companies only require the name of the chemical before it arrives, and require that the MSDS accompany the shipment and be installed near the point-of-use before the chemical is delivered to the fab area.

Even when the MSDS is obtained before the shipment arrives, sometimes additional information is necessary to determine if controls are adequate and/or to determine if air sampling should be performed. The first step is to consult established references such as Patty's *Industrial Hygiene and Toxicology*^[3] and the American Conference of Governmental Industrial Hygienist's (ACGIH) *Documentation of the TLV's*.^[4] When

information from the MSDS and the standard references is lacking, a computer search of the published literature is needed.

These searches can be performed directly through organizations that maintain individual databases such as those provided through the National Library of Medicine Toxicology Information Program Online Services, (301) 496-1131, or indirectly through commercial information services such as Dialog Information Services, (800) 334-2564. The main databases for toxicology information include TOXLINE, EMBASE, MEDLINE, and CA SEARCH. Searches of these databases can be done on-line or, in most cases, by purchasing CD ROMs containing the information that is often updated on a quarterly basis. The CD-ROM option is becoming more popular due to decreasing costs, and this medium allows the manipulation of the information in electronic form.

To ensure complete and up-to-date information, the best approach is to use an on-line commercial service. Most mid-sized and larger semiconductor companies will already have an account with a commercial service through the company's information center. Alternatively, with a PC, modem, and a four- to eight-hour introductory course, the end user can perform the searches directly.

If the MSDS, reference books, and results of the literature search still leave major uncertainties in the information necessary for the initial evaluation, the research department for the chemical manufacturer can sometimes help fill these gaps or provide the names of other manufacturers who are using the chemical.

As with all routine occupational industrial hygiene monitoring for assessing semiconductor processes, it is necessary to have a list of the chemicals and radiation sources used in the processes and to know the appropriate monitoring methods. If an appropriate method is not available, the industrial hygienist will need to work closely with the analytical laboratory that will be performing the sample analysis. This can be a very time-consuming process and caution should be used in balancing the severity of the potential hazard versus the cost of method development and analytical testing.

In addition to the parent chemistries, semiconductor processes also may involve reaction products that can create significant exposures. The classic example is the use of arsine gas in an implanter which can result in possible arsenic exposure for maintenance workers.^[5] Many of the references cited in this chapter provide additional insight into the potential hazards that may be encountered from reaction products, physical agents, and process effluent treatment systems.

Radio frequency (RF)/microwave radiation is a good example of a potential hazard that does not have good warning properties. Therefore, baseline data should be obtained for all routine operations with a potential for RF/microwave radiation exposure, and baseline RF surveys of new equipment may be required. As a practical matter, a lower frequency limit needs to be drawn to assess when baseline surveys are necessary. Since the body is fairly transparent to RF frequencies in the kilohertz region, and standard RF/microwave radiation meters have a lower limit of 300 or 500 kHz (depending on the meter), 500 kHz is sometimes used as a cutoff for the lower limit for thermal effects. Most RF/microwave equipment used in semiconductor manufacturing, such as plasma etchers and ashers, sputtering units, mold pre-heaters, microwave ovens, and plasma enhanced CVD units, operates at or above a frequency of 13.56 MHz. Table 3.11 in Ch. 3 (Physical Agents) provides additional information on the RF/microwave frequencies that are used on common semiconductor equipment.

Another area that is important to investigate is exposure during maintenance of RF equipment, when by-passing the shielding and/or internal grounding may greatly affect the emission levels radiating from the machine. If the appropriate shielding and internal grounding is in place, routine exposures to semiconductor fab operators should be very low.

When characterizing employee exposures to physical agents such as electromagnetic fields (EMF) which include radio frequency (RF), microwave radiation, and ELF (*extremely low frequency*)/VLF (*very low frequency*), direct reading meters for RF monitoring are more easily used for determining emissions from the fab equipment than they are for estimating dose. Therefore, RF surveys within the semiconductor industry often focus more on equipment emissions than they do on estimating the person's actual exposure, however this may change if two IEEE standards recently approved by ANSI are implemented by the industry.^{[6][7]} The focus on equipment also applies to potential exposures from other non-ionizing radiation sources (e.g., ultraviolet, laser, EMF, etc.).

Ideally, companies should have baseline industrial hygiene data on all routine operations with a potential for chemical, RF/microwave radiation, or ionizing radiation exposures. Even when the industrial hygienist believes the exposure will be well below the allowable limit for an operation, monitoring data is beneficial because:

- Data provides objective documentation on exposures from the operation rather than a qualitative evaluation

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- During the monitoring survey, a detailed evaluation of work practices should be made, and when appropriate, modifications initiated
- During the survey, non-routine operations which may be performed can be identified and a determination made if additional evaluations are necessary
- Data provides a quantitative measure that will assist with determining if medical monitoring is appropriate
- In the event of litigation, the data will aid in establishing actual exposure levels for employees involved in the operation

1.2 **When to Re-monitor**

As a minimum, an operation should be re-sampled if it is moved or modified or there is other reason to suspect the exposure may have increased. Compounds that OSHA requires initial sampling for also have a provision for resampling (e.g., arsenic and lead) when the exposure exceeds the *action level* for the chemical.

Because of the dynamic nature of semiconductor operations, it is not always possible to know when an operation has been modified. Changes in work practices can also cause increased exposures. Therefore, operations should be periodically reassessed. The resampling frequency should be based on the likelihood of a significant exposure. Some semiconductor companies have internal guidelines for resampling frequencies.^[5]

1.3 **When Monitoring is Not a Priority**

If the exposure is going to be reduced regardless of any air sample results (e.g., odor complaints, prudent avoidance, or unacceptable exposure), consideration should be given to the value of performing the IH sampling. The rule of thumb is: *if the data does not help to protect the employee or the company, do not sample*. Examples:

- When responding to an odor complaint from a photoresist spinner operation, find and mitigate the cause of the odor rather than expend the time and resources needed to perform sampling
- If the operators are cleaning work surfaces with methanol consider substituting water, ethanol or isopropanol, or mixtures of these materials

- If a maintenance technician is sticking his head inside the open-top vapor degreaser to clean it, modify the operation to eliminate the exposure

2.0 ASSESSMENT TOOLS

It is not always practical to have baseline industrial hygiene data on all routine operations. In this case, decisions must be made on where to spend the available time and money. The first step in the decision process is to conduct an exposure assessment of all operations involving potential chemical or radiation exposure. The exposure assessment should include: reviewing operating procedures and maintenance records; observing the operation; and discussing it with the operator, equipment technician, process engineer and the supervisor. Results of the exposure assessment should be documented, particularly when the decision is made not to monitor. Assume this decision may become the focus of future litigation and document it carefully.

All chemicals and radiation sources need to be identified during the exposure assessment. Some semiconductor manufacturing operations involve reaction products that are formed by the process (e.g., plasma systems, chemical vapor deposition systems, and ion implanters). When the composition of the reaction products is uncertain, bulk samples are taken and analyzed for specific compounds identified in previous surveys and then for unknowns. Data from the bulk samples are used to determine if subsequent personal breathing zone samples are needed.

2.1 Exposure Assessment

One methodology that has been proposed by the Organizational Resource Counselors (ORC) for performing generic workplace exposure assessments is explained in the form of two hypothetical exposure assessments.^[8] The first example involves evaluating a homogeneous exposure group (HEG) for acid etch operators. The second hypothetical example is an exposure assessment involving wave solder maintenance.

Both examples start with a qualitative assessment. Based on this, the determination is made as to whether or not a quantitative assessment (i.e., IH monitoring) is needed. While this approach relies on professional judgment that may not always be the strongest defense in the event of litigation, it does optimize time and money expenditures to best protect the employee.

Hypothetical Qualitative Assessment - Acid Etch Operators

Name of HEG or Operation: Acid Etch Operators.

Location: Bldg. 5 upper (near columns D53 and D55).

Hazardous Chemicals Involved:

Mixed Acid Etch (Nitric Acid, 55%; Acetic Acid, 20%; Hydrofluoric Acid, 10%; Water, 15%). Located in Bay 5.

Stress Relief Etch (Nitric Acid, 75%; Acetic Acid, 12%; Hydrofluoric Acid, 7%; Water, 6%). Located in Bay 5.

Quantities of Materials Used: 10 liter etch baths (usually filled about 75% full); four baths total (two of each etch solution).

Description of Operation: Wafer fab operators dip wafers in etch baths and allow wafers to remain in the baths for predetermined time (2 to 10) minutes; at the end of the time, the wafers are removed and rinsed in DI water baths and removed; operators top-off baths at end of day and make-up new baths at the start of the week.

Number of Persons: Day shift only; six wafer fab operators (four use stations more than other two).

Potential Route(s) of Exposure:	Inhalation	Ingestion	Skin Contact
	[X]	[]	[X]

Frequency and Variability of Exposure: Great variability in usage depending on wafer fab operator and work load; average about six runs per day per bath; maximum usage about 12 runs per bath per day. Also, top-off acid baths at end of shift and usually make-up baths at start of week by aspirating old bath and hand pouring new ones.

Types and Variability of Tasks: Other operations performed by wafer fab operators include set-up of wafers, use of drying ovens, occasional microscope work (inspection), and logging data on terminal. Aside from etching, the wafer fab operators do not work with other chemicals. On the average, etch operations occupy roughly one-third of their shift.

Controls: Wet decks are provided with exhausted enclosures with average face velocities of 110–160 linear feet per minute. Each wet deck exhaust has a magnehelic with a set-point. The safety committee formally checks the magnehelic once a month and operators say they normally check it every morning. The wet decks are rinsed down at the end of the shift. Opening and pouring of the acid bottles are done only in the exhausted enclosure; the operators wear aprons, arm guards, face shields, safety glasses, and gloves (Acme latex acid gloves). PPE is not shared between fab operators; gloves are changed at least once per shift. The standard operating procedure for both etch operations contain the safety precautions for the processes. Operators were trained on standard operating procedures (SOP) and received chemical safety training.

**Hypothetical Qualitative Assessment
Acid Etch Operators
(continued)**

Employee Comments: Talked to four of the six wafer fab operators—Paula Garcia, Joe Ng, Sam Smith, Jane Wong (Karen Reynolds was in training and Pat Nelson was on vacation). None of the wafer fab operators had safety concerns related to the etch baths. None of the operators reported smelling or sensing acids during any part of the process. The supervisor, Pam Prescott, and the technician, John Kim, said they also were not aware of any problems with the etch process. John Kim (who had been in the department the longest, since 1982) said there had been only one spill in the area since he started, it was in 1987; a one gallon bottle of Mixed Acid Etch tipped over on the deck and about half a bottle spilled on the floor; no one was splashed; the area was evacuated and the ERT called; no abnormalities in clean-up were mentioned.

General Observations: Good housekeeping in the area, secondary containment and chemical storage cabinets do not show signs of old leaks or spills. Zeroed magnehelics; both were O.K. Observed the manual top-off of baths; it was done according to the SOP; no odors were observed.

Past Exposure Monitoring Data or Health Surveillance Data: None available

Qualitative Estimation of Exposure Level: Peak exposures during normal operations are less than the odor threshold for these mixtures (0.3 ppm as nitric acid and about 0.08 ppm as acetic acid).

Quantitative Assessment Needed?: No

Additional controls needed?: None needed

Re-evaluation Frequency: Once every three years—unless there is a change in the operation, the wafer fab etch operator HEG will be re-evaluated in Q4 1996.

Other Comments: Pam Prescott said there were very tentative plans to partially automate the operation next year. She will call if the decision is made to make the changes.

Industrial Hygienist:
[Signature]:

Knowledgeable EHS Person: Not Applicable

Date:

**Hypothetical Qualitative Assessment
Wave Solder Maintenance Operation**

Name of HEG or Operation: Wave Solder Maintenance Operation

Location: Electrovert Ultra, 3 lower (near column G13); Takashimatta, 2 upper (near column B3); Kennapor, 2 upper (near column C9).

Hazardous Chemicals Involved: Kester 63/37 solder, solder dross, and resin flux (mixture of alcohols and terpene-based resins)

Quantities of Materials Used: About five kg of solder is used in each machine; about 100 gm of solder/solder dross remains on the solder pump that is removed and cleaned during maintenance. Virgin flux is dispensed from 5 gallon containers. Waste flux is drained from the flux reservoir into five gallon chemical waste containers.

Description of Operation: The maintenance technicians remove and clean the solder pumps in accordance with the Standard Operating Procedure for the operation . The pumps are cleaned on a work bench in the tech area by scraping with a metal spatula and brushing with a wire brush.

Number of Persons: Two technicians perform the operation.

Potential Route(s) of Exposure:	Inhalation	Ingestion	Skin Contact
	[X]	[X]	[X]

Frequency and Variability of Exposure: Each machine is serviced once a month on different days. The Ultra is used more than the other two and has a greater buildup of the Sn/Pb residues, and therefore, would represent the greatest exposure. The actual cleaning of the pump (scraping and brushing) takes about one hour.

Types and Variability of Tasks: No other operations are performed by the technicians that involve potential tin or lead exposure.

Controls in Use: During the cleaning of the solder pump, the technician wears chemical gloves (latex acid gloves), and safety glasses. After the pump is cleaned, the top of the workbench is cleaned with a cleaning solution (409 Cleaner). Gloves are disposed of at the end of the operation. The standard operating procedure for the wave solder maintenance contains the safety precautions for the process. The technicians have received lead safety training and participate in annual health surveillance for their lead exposure. The technicians routinely wash their hands before leaving the area.

**Hypothetical Qualitative Assessment
Wave Solder Maintenance Operation
(Continued)**

Employee Comments: Talked to both technicians—Mike West and Jamie Martinez. Neither had concerns regarding the operation. They said their health surveillance data (blood lead monitoring) did not indicate problems. When asked about eating, drinking, and smoking in the area, they said they were not allowed to eat or smoke in the area, and drinking was only allowed at their desks.

General Observations: Good housekeeping in the area, no visible indications of surface contamination. No signs of smoking or eating. Two desks had coffee cups. Scrapers and wire brushes not cleaned after use, but they are only used for the pump clean and they are kept in a small metal cabinet under the work bench that is dedicated to their storage.

Past Exposure Monitoring Data or Health Surveillance Data: No exposure monitoring data available. Health surveillance data negative.

Qualitative Estimation of Exposure Level: Based on the technicians' description of the operation and exposure data from other similar maintenance operations, the levels probably exceed 10% of the PEL/TLV.

Quantitative Assessment Needed?: Yes.

Additional Controls Needed?: Yes; both technicians need to enter the respirator program (training, fit testing, and health surveillance) before the next wave solder maintenance and use air purifying respirators during the operation. Pending the outcome of air sampling, local exhaust ventilation may be needed during the pump cleaning. The metal cabinet used to store scrapers and wire brushes should be labeled to warn technicians that the tools are contaminated with lead and should not be used for purposes other than wave solder maintenance. Also, a dedicated HEPA vacuum should be provided for the cleanup of dross-contaminated residues in the area and on equipment surfaces.

Re-evaluation Frequency: Pending outcome of air sampling.

Other Comments: None

Industrial Hygienist:

[Signature]

Knowledgeable EHS Person: Not Applicable

Date:

2.2 **Monitoring Plan**

Identifying routine cleanroom operations with potentially significant exposures is normally a simpler task than identifying potentially hazardous maintenance operations. Walk-throughs of the area and discussions with the operators are usually enough to determine what operations need sampling. Even non-routine exposures to chemicals are often quickly identified through monitoring alarms for toxic gases with poor warning properties. Cleanroom personnel will usually mention new or unusual odors in their area.

Maintenance operations are usually more difficult to identify, and in semiconductor manufacturing, it is the maintenance operations that have the greatest potential for high exposures.^{[9]-[13]} Information on maintenance operations comes from talking with engineers and technicians who perform the operations, and reviewing maintenance specifications. Formal surveys of the technicians can also be useful in identifying these operations. A hypothetical monitoring strategy for a wave solder maintenance operation is presented to highlight the pertinent topics that need to be covered.

The actual establishment of the monitoring plan involves the gathering of numerous pertinent pieces of data that will guide the industrial hygienist. Some of the elements that need to be identified are:

- The jobs or tasks to be sampled
- The specific chemical or biological agents to be evaluated
- The number of samples to be taken
- The frequency of monitoring
- The nature of the sample to be taken (i.e., personal vs. full-shift TWA vs. STEL, etc.)
- The specific instruments to be used
- The method of calibration of the instrument or sampling train
- Assuring “blanks” of the sampling media and sample containers are submitted with the samples
- Pertinent ambient conditions are noted (i.e., temperature, atmospheric pressure, relative humidity, air flow patterns, etc.)
- The types, models and conditions of personal protective equipment in use
- The routes of exposure to the employees for the agents being monitored

Hypothetical Monitoring Strategy Wave Solder Maintenance Example

Jobs, Tasks and/or Areas to be Sampled: Wave Solder Maintenance. Will sample pump maintenance on the Ultra in 3 lower (near column G13).

Chemical or Biological Agents: While the solder contains both tin and lead (ratio 63/37), as a percentage of the PEL/TLV lead will be the limiting factor. This sampling will focus on the lead exposures only because its toxicity and lower PEL/TLV. Subsequent sampling may also include monitoring for tin and alcohol-based flux.

Number of Samples, Frequency of Monitoring, and Nature of Samples (e.g., personal vs. area, full-shift TWA vs. STEL, etc.):

Personal air samples will be taken for both technicians. Two cassettes will be taken for each tech; one while the pump is being cleaned on the work bench, and another during other operations where the exposures are not expected to be significant (e.g., pump removal and re-installation).

An area air sample will be taken at the work area adjacent to the work bench during the cleaning of the pump to document potential worst case exposure to persons not involved in the maintenance.

Wipe samples will be taken: on the work bench before pump maintenance (2 wipes) and post-clean after pump maintenance (5 wipes); on the wave solder machine at one or two locations before it is opened for maintenance (2–4 wipes) and after maintenance is complete (2–4 wipes); on the desk closest to the work bench (3 wipes); on the fingers of the technicians after removing gloves and washing hands (2–4 wipes).

Additional Notes: On the day of the sampling, IH will talk with the maintenance supervisor; if he wants, IH will take two additional wipe samples in the metal cabinet used to store the tools. On the day of sampling, IH will also evaluate the potential for tracking solder from either the wave solder area or the maintenance area to the hallways (additional wipe samples may be needed).

Industrial Hygienist:

[Signature]

Date:

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- The characterization of the engineering controls that are relevant to the exposure to be monitored (i.e., local exhaust hoods, enclosures, glove boxes, lab hoods, HEPA vacuums, etc.)
- Photographs and diagrams of the work area and sampling locations

Once the monitoring has been performed, samples must be handled carefully, submitted to a qualified analytical laboratory, and results calculated once the values are reported from the lab. In some cases, the use of biological monitoring data may be useful as another datapoint in the determination of the hazard of an operation or task.

Routine biological monitoring (e.g., arsenic levels in urine and lead levels in blood) is used by some companies to supplement air sampling results. The greatest advantage of biological monitoring is that the biological parameters of exposure takes into consideration absorption by routes other than the lungs. However, a variety of conditions and practical considerations limit the effectiveness of biological monitoring.^[14] Based on data from extensive past biological monitoring within the industry, urinary arsenic testing for silicon-based processing is of limited value. Its primary utility continues to be associated with gallium arsenide-based processing, particularly in maintenance activities and with certain wave-solder maintenance operations. A detailed discussion of the use of urinary arsenic testing will be provided to illustrate the use of biological monitoring in semiconductor industrial hygiene assessments.

3.0 BIOMONITORING: URINARY ARSENIC TESTING

The use of urinary arsenic monitoring will be discussed as an example of a biomonitoring technique that can be used as a supplement to air sampling for evaluating routine arsenic exposures. Urinary arsenic monitoring is particularly useful for determining if ingestion of arsenic from improper personal hygiene is occurring. It may also be used in identifying improper respirator usage, and for providing information on unanticipated exposure. Biological monitoring is also important for estimating the extent of exposure during an accidental exposures such as an arsenic release. Often, it is the best method for establishing exposure levels in these cases.

Severe acute arsenic poisoning from semiconductor manufacturing operations is an extremely rare event. Treatment for severe arsine exposure would be based on elevated serum hemoglobin or the occurrence of oliguria rather than a specific exposure level.^[15] Urinary arsenic data is primarily used for documentation purposes and to trigger further investigation.

3.1 Who Should be Sampled

Urinary arsenic samples should be taken for persons working in arsenic usage or equipment maintenance areas where their 8-hour *time weighted average* (TWA) airborne arsenic exposure, without regard to the use of a respirator, exceeds $5 \mu\text{g}/\text{m}^3$ at least 30 days per year.^[16] Urinary arsenic samples may also be included:

- as a periodic check for arsenic ingestion
- as part of a respirator program evaluation
- as part of an initial evaluation of an operation

Samples should also be taken when it is reasonable to expect a person may have been exposed to significant amounts of arsenic during an accidental arsenic/arsine release. Normally, all exposed persons in the worst-case group should be sampled.

3.2 Sampling Frequency

Yearly urinary arsenic analysis should normally be adequate. Semi-annual urinalysis may be reasonable in instances where semi-annual medical surveillance is required by the OSHA arsenic standard (e.g., persons 45 or older with at least 10 years exposure over the arsenic action limit).^[15]

The half-life of inorganic arsenic in the body is listed in some references as 10 to 59 hours.^{[17][18]} The ACGIH assumes pulmonary clearance of arsenic particulates undergoes a triphasic elimination with half-lives of four days for the first phase and ten days for the second, with a very small third phase being retained in the body for years.^[19]

For an accidental exposure, urinary arsenic samples should be taken as soon as practical after the incident. Follow-up monitoring is recommended within 24 hours of the incident, with a “first void” method being commonly used. If the data show elevated urinary arsenic levels, additional samples should be taken until levels return to background concentrations. For routine biological monitoring, the samples should be taken at the end of the workweek, with early morning sampling at home to avoid contamination.^{[19][20]} Prior to the collection of samples, the analytical laboratory performing analysis should be contacted to determine specific sample handling recommendations and if refrigeration of collected samples is needed.

3.3 Instructions to Employees

The following are typical of the instructions given to employees who submit urine samples for analysis:

1. Do not eat fish or shellfish for five days before your test (i.e., any fish, clams, shrimp, lobster, prawns, calamari, etc.). These foods contain organic arsenic that can increase the results of your test.
2. Do not drink any red wine for five days before your test; some red wines contain low levels of arsenic.
3. Do not smoke three days prior to your test. (Note: at least one semiconductor manufacturer excludes smokers from routine arsenic testing).
4. Wash your hands before urinating.
5. Urinate in the container that was given to you immediately after waking up in the morning.
6. Close the container tightly and bring it with you to your appointment.
7. Collect the specimen only if you have not eaten seafood or drank red wine within five days of your test.

Arsenic levels greater than 1000 $\mu\text{g/l}$ can occur when certain species of fish are eaten.^[21] This makes abstinence from fish prior to taking the sample particularly important. The five day fish restriction should be considered a minimum. Organically bound arsenic from fish, primarily arsenobetaine, is rapidly excreted from the urine with a half-life of 12 to 24 hours, and causes no known toxic effects.^{[22][23]} Assuming a conservative example: 1000 $\mu\text{g/l}$ of arsenobetaine in a person would be decreased to 31 $\mu\text{g/l}$ after five days of abstinence from fish. After seven days, the level would be less than 8 $\mu\text{g/l}$. When diet is not a factor, urinary arsenic levels in the general population are normally 5 to 30 $\mu\text{g/l}$.^[21]

The restriction on red wine consumption may be more restrictive than necessary. Arsenate insecticides are no longer used in the U.S., and are seldom used abroad. However, the suggested international limit for arsenic in wine is 200 $\mu\text{g/l}$.^[24] This relatively high limit for arsenic in wine may justify the restriction.

Another potential source of arsenic is tobacco smoking. Approximately 12–42 μg of arsenic is contained in each cigarette.^[25]

3.4 Urinary Arsenic Sample Analysis

Atomic absorption (AA) spectrophotometry is preferred over colorimetry for urinary arsenic analysis.^[26] The preferred method for total urinary arsenic is wet digestion with HNO_3 , H_2SO_4 , and $\text{K}_2\text{Cr}_2\text{O}_7$ and direct hydride AA.^[27]

Currently, there is no independent quality assurance program for laboratories performing urinary arsenic analysis. Also, some semiconductor manufacturers have found poor recoveries from blind spiked samples sent to a lab as a quality control check.^[28]

A recommended quality assurance program for arsenic urine analysis labs should include spiked samples submitted for the program which contain trivalent inorganic arsenic, pentavalent inorganic arsenic, monomethylarsonic acid, and dimethylarsinic acid. However, finding a commercial source of arsenic reference standards is difficult.

3.5 Data Interpretation

In the past, labs performing urinary arsenic biomonitoring recommended action levels of approximately 80 to 100 $\mu\text{g/l}$. The ACGIH has established a Biological Exposure Indices (BEI) of 50 $\mu\text{g/g}$ creatinine for arsenic.^[19] Spot urine samples average about 1.4 grams of creatinine per liter of urine.^[29] Therefore, the BEI for arsenic is calculated to approximately equal 70 $\mu\text{g/l}$. Irrespective of these values, when a routine exposure to inorganic arsenic is suspected, investigation and possibly resampling should be considered if dietary sources of arsenic are minimized and urinary arsenic levels exceed 30 $\mu\text{g/l}$.

When diet is not a factor, urinary arsenic levels in the general population are normally 5 to 30 $\mu\text{g/l}$.^[21] In one study, the correlation of 8-hour average airborne exposures to inorganic arsenic and urinary arsenic levels showed a regression equation of:

$$y = 0.304 x$$

where y equals airborne arsenic in $\mu\text{g/m}^3$ and x equals urinary arsenic levels, corrected to a specific gravity of 1.018, in $\mu\text{g/l}$.^[30]

Applying this equation to the urinary arsenic levels in the general population, it results in equivalent airborne levels of 1.5 to 9.1 $\mu\text{g/m}^3$. This indicates that continuous airborne exposures to arsenic at OSHA's action level (5 $\mu\text{g/m}^3$) are difficult to distinguish from background urinary arsenic levels, even when diet is not a factor.

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Another study showed a different relationship between urinary arsenic levels and arsine exposures.^[31] The correlation between airborne arsine and urinary arsenic can be expressed by the following equation:

$$y = 2.43 x + 11.99$$

where y equals urinary arsenic in $\mu\text{g/l}$ and x equals airborne arsine levels in $\mu\text{g/m}^3$. Using this equation, an 8-hour exposure to $5.0 \mu\text{g/m}^3$ of arsenic would result in a urinary concentration of $24 \mu\text{g/l}$.

Differences in the two studies may be accounted for by differences in biotransformation of the inorganic arsenic species or by differences in study methodologies. However, they both indicate urinary arsenic levels above background levels may be significant. A more thorough review of the relationship between arsenic intake and urinary excretion is covered in Refs. 19 and 20.

Biological monitoring is commonly performed when exposure to other compounds such as antimony and lead occur. A study was performed of liver enzyme testing among microelectronics equipment maintenance technicians, and found a possible connection between alcohol consumption and occupation, and abnormal liver enzyme results.^[32]

4.0 CHEMICAL AGENTS

Most chemical exposures during routine semiconductor manufacturing operations are far below accepted exposure standards.^{[9][11][12][33][34]} Exposures presented in Table 2.1 are only for published data on semiconductor manufacturing operations that have shown significant airborne concentrations during atypical operations or maintenance activities. This information is intended to aid industrial hygienists and safety engineers in recognizing potential exposures that may require particular attention.

4.1 Air Contaminants

Table 2.1 lists semiconductor manufacturing operations that can result in significant chemical emissions during atypical operations or maintenance activities. The exposure levels listed in Table 2.1 are the maximum values cited in the references that potentially relate to an employee exposure. They often represent less than an eight-hour exposure. Most of the exposures are the result of improper work practices and are not levels typically found in the semiconductor industry.

Table 2.1. Semiconductor Manufacturing Operations with Potentially Significant Chemical Exposures

Process	Activity/Equipment	Substance	Maximum Exposure*	(Ref.)
GaAs & InP Wafer Manufacturing	Weighing Arsenic	Arsenic	Not Listed	[11]
	Loading Furnace	Arsenic	14 µg/m ³	[13]
	Cleaning Crystal Growers	Arsenic	2700 µg/m [35]–[37]	
	Bead-Blasting GaAs Ingots	Arsenic	1500 µg/m ³	[13]
	Grinding/Sawing Ingots	Arsenic	7.2 µg/m ³	[13]
	Sawing GaAs Ingots	Arsine	92 ppb	[38]
	Sawing InP Ingots	Phosphine	176 ppb	[38]
	Cleaning Work Surfaces	Arsenic	Not Listed	[11]
Wafer Preparation	Washing Silicon Wafers	Methanol	931 ppm	[9]
	Stripping Carrier Pads	Methylene Chloride	522 ppm	[9]
Photolithography	Resist Filter Maintenance.	EGMEA	6.3 ppm	[13]
Etching	Polysilicon Batch Etching	Fluorides	Not Listed	[11]
	Plasma Al Etcher Maint.	Hexachloroethane	Not Listed	[39]
	Plasma Al Etcher Maint.	Cyanogen Chloride	3.3 ppm	[40]
	Plasma Al Etcher Maint.	Hydrogen Chloride	68 ppm	[13]
Diffusion and Deposition	Furnace Maintenance	Arsenic	85 µg/m ³	[9]
Ion Implantation	II Source Housing Maint.	Arsenic	3200 µg/m ³	[41][42]
	Annual PM	Arsine	2,200 ppb	[43]
	Quarterly PM	Arsine	>2000 ppb	[43]
	II Manipulator Maint.	Phosphine	1725 ppb	[49]
	II Cryo-Pump Maint.	Arsine/Phosphine	0.06 ppm	[13]
	HEPA Vacuum Maint.	Arsenic	>1000 µg/m ³	[13]
Chemical Vapor Deposition	Epi Prescrubber Maint.	Hydrogen Chloride	>5 ppm	[11]
	GaP CVD	Prescrubber Maint.	>0.2 ppm	[11]
	GaAs CVD Prescrubber Maint.	Arsine	>0.05 ppm	
[11]				
	GaAs MBE Reaction Chamber Maint.	Arsine	0.08 ppm	[38][44]
Metallization	Evaporator Bell Jar Maint.	Silver	>100 µg/m ³	[11]
	Evaporator Bell Jar Maint.	Methyl Alcohol	>250 ppm	[11]
Assembly and Test	Wave Solder Maint.	Lead	>50 µg/m ³	[13]
	Solder Pot Maint.	Lead	>50 µg/m ³	[11]
	Degreaser Maint.	Degreasing Solvents	Not Listed	[11]
	Marking Test	Monoethanolamine	> 3.0 ppm	[45]

* Exposures listed are the maximum levels reported in the references cited that could represent potential employee exposures. Typically, they are for less than an eight-hour exposure and do not take into consideration possible respirator usage. Also, many of the exposures are the result of improper maintenance techniques or procedures.

Note: A process description for each of the major semiconductor device types are provided in: Appendix A: Silicon Ingot and Wafer Manufacturing, Appendix B: Silicon Device Manufacturing, Appendix C: III-V Ingot and Wafer: Light Emitting Diodes (LED), and Appendix D: III-V Microwave IC.

Process: Wafer Manufacturing and Preparation. Gallium arsenide wafer manufacturing represents the semiconductor manufacturing process with the greatest potential for significant, routine chemical exposures. While gallium arsenide wafer manufacturing is done only by a small percentage of semiconductor manufacturers, particular emphasis is needed in this area. The large amounts of arsenic used in the process, the numerous steps in the process, and the low airborne exposure limit for arsenic make it difficult to control exposures. Articles by Harrison,^[46] Lenihan et al.,^[35] McIntyre and Sherin,^[47] and Sheehy and Jones^[48] cover hazards and controls for this process. Arsine and phosphine may be formed from the reaction of gallium arsenide and indium phosphide with moisture in the air, other arsenides and phosphides, or when mixed with acids during the processing of gallium arsenide and indium phosphide.^{[38][49]} In these same papers, the authors also reported 92 ppb arsine and 176 ppb phosphine two inches away from the slicing blades used to cut gallium arsenide and indium phosphide ingots.

In silicon wafer manufacturing, solvent exposure during silicon wafer preparation can be hazardous if *local exhaust ventilation* (LEV) is not used. Methanol exposures of up to 931 ppm were reported during the washing of silicon wafers without adequate LEV. Methylene chloride exposures of up to 522 ppm were reported when no local exhaust was used during the stripping of carrier pads from their metal carrier plates.^[9]

Process: Lithography. Lithography—also referred to as photolithography, microlithography, photomasking, and masking—is a crucial step in the manufacturing process. Through a photography-like process, precise patterns are formed on the wafer. The key to forming these patterns is transferring the image from a mask to a radiation-sensitive chemical mixture called a resist (see Appendix B for a process description and Ref. 52 for process details).

Resists are comprised of several components. The classic composition of negative photoresist is an azide-sensitized rubber such as a polyisoprene polymer base with a bis-aryldiazide sensitizer, and a carrier solvent that makes the resist a liquid, allowing it to be spun on the wafer as a thin layer (sometimes additives are included such as dyes to control the light rays). A typical positive photoresist would be a phenol-formaldehyde novolak resin structure and a diazonaphthoquinone sensitizer in a carrier solvent.

Exposure systems for most resists use ultraviolet (UV) light from mercury-vapor arc lamps. The resists are designed to react either to a broad spectrum of UV wavelengths, or they are formulated to react

preferentially to one or more of the main spectrum lines emitted from the lamp (e.g., g-line at 435 nm; h-line at 405 nm; and i-line at 365 nm).

The use of resists that utilize lower UV wavelengths is a growing trend in the semiconductor industry due to the requirements for smaller linewidth geometries. The practical resolution limit for conventional lithography is slightly greater than the wavelength of light used.^[50] When wavelengths less than 365 nanometers (nm) are needed, exposure sources other than mercury-vapor arc lamps are used. By increasing resolution, these are:

- Deep UV systems (e.g., excimer laser sources at 248 nm wavelength)
- Electron beam systems
- X-ray sources (typically 0.4–5.0 nm wavelength)

Table 2.2 is a compilation of information regarding some commercially available resist systems.

In terms of routine chemical exposures, photolithography is the semiconductor manufacturing process that generates the greatest concerns. From an IH perspective, the main focus is on the carrier solvent. Vapor pressures for the base materials, sensitizers, and additives used in the resists are very low.

Concerns in lithography are primarily caused by potential reproductive hazards associated with selected positive resists (e.g., ethylene glycol monoethyl ether acetate as a carrier) that are currently being phased out by the industry. Occasional odors from the negative resists (e.g., xylene) also result in employee concerns. Because of these concerns, semiconductor industry industrial hygienists spend a great deal of time sampling photoresist operations. While this is useful in characterizing these operations, routine exposures during spinner and developer operations are typically less than five percent of the TLV/PELs for the solvents used in the process.^[33]

A one-hour exposure to ethylene glycol monoethyl ether acetate (EGMEA) of 6.3 ppm was found during the use of a spinner dispense system. This exposure was primarily caused by poor work practices during the maintenance operation.^[13]

Process: Etching. Etching is a part of the overall process of lithography. During this step, the pattern formed on the wafer by the resist is permanently etched into the wafer by the use of wet chemicals or dry chemical techniques (e.g., plasma or reactive ion etching). After the pattern is etched into the wafer, resist is removed (stripped) from the wafer. This is also done by using wet chemical or dry chemical (e.g., plasma ashing) procedures.

Table 2.2. Typical Resist Systems Used in Photolithography.^[51]

Company Name: Resist Designation	Type	Intended for exposure by:							Resolution Limit (μm)	Solids Content (Weight %)	Solvent Type	Flash Point (C.C. °C)
		B	g	h	i	DUV	EB	XRAY				
Gensis GS-7010	P	*	*	*	*				0.25	15-35	EL	--
Hoechst Celanese Corp.												
AZ1500 Series	P	*	*	*					0.7	26-30	PGMEA	47
AZ1600 Series	P	*	*						0.5	26-30	PGMEA	47
AZ6200B Series	P		*						0.45	26-30	PGMEA	47
AZ7500 Series	P				*				<0.35	26-30	PGMEA	47
AZ7500 T Series	P				*				0.75	32-38	PGMEA	47
AZ DX Series	P					*			0.225	20	PGMEA	47
AZ P4000	P	*	*						0.8	25-47	PGMEA	47
AZ PF 500	P						*	*	<0.20	28	PGMEA	47
IBM, Tech. Prod. Div. APEX	P					*			0.225	15-25	PGMEA	47
Morton Electronic Materials												
EL2015	P	*	*	*	*				0.5	24-30	EL	52
EL2025/26/26TE	P								0.5	36-50	EL	52
EPA914 EZ	P	*	*	*					0.6	21-40	PGMEA	42
NOVA 2050 AR-2	P	*	*	*					0.6	18-42	PGMEA	42
NOVA 2070	P	*	*	*	*				0.35	24-40	EL	52
NOVA 2071	P	*	*	*	*				0.35	24-40	EL	52
OFFPR-800	P	*	*	*					1.0	18-37	EEA	51
OFFPR-800PG	P	*	*	*					1.0	24-32	PGMEA	42
OMR-83	N	*			*				2.5	11-14	Xylene	28
PR1024 MB	P				*	*		*	0.35	12-19	EEA	51
OCG Microelectronic Materials Inc.												
CAMP-6 Series	P			*	*	*			0.25	14-17	EEP	--
HiPR 6500 Series	P	*	*						0.4	23-32	EL/EEP	--
HPR 500 Series	P	*	*						0.8	21-37	EL	--
OCG 825 Series	P	*	*	*					0.6	29-40	EEP	--
OCG 895 Series	P				*				0.4	18-40	PGMEA	--
OCG 897 Series	P				*				0.35	various	PGMEA/EEP	--
OiR 32 Series	P				*				0.32	various	MMP	--

Table 2.2. (Cont'd)

Company Name: Resist Designation	Type	Intended for exposure by:							Resolution Limit (μm)	Solids Content (Weight %)	Solvent Type	Flash Point (C.C. °C)
Shipley Company Inc.												
Megaposit S1400	P		*						0.6	17-34	EEA	41
Megaposit S1800	P		*						0.6	18-33	PGMEA	46
Megaposit S3800	P		*	*	*				0.45	21-27	EL	39
Microposit SAL601	N						*		<0.1	10-29	EEA	41
Megaposit SPR2	P		*						0.4	22-29	EL	39
Megaposit SPR500-A	P				*				0.35	16-32	EL	40.5
Megaposit SPRT500-A	P				*				0.35	16-32	EL	40.5
System 8	P		*						0.5	10-31	EL	41.1
Megaposit XP-89131	N					*			0.225	16-22	PGMEA/Anisol	43.3
Microposit SAL603	N						*		<0.1	10-29	PGMEA	41
Microposit SAL605	N						*		<0.1	10-29	PGMEA	41
Megaposit XP-90236	N					*			0.25	18-22	PGMEA/Anisol	43.3
Toray Industries, Inc.												
Photomec Series	N	*	*						2	16-18	NMP	60
EBR-9	P						*		0.2	6.1-6.2	MEA	29.5
UCB-JSR Electronics, Inc.												
PFR GX210ED2	P		*						0.58	--	EL	46
PFR GX250EL	P		*						0.42	--	EL	46
PFR IX405EM	P				*				0.40	--	MMP/EL	46
PFR IX500EL	P				*				0.34	--	EL	46
PFR IX560FDM2	P				*				0.36	--	EL/MMP	46
PFR IX700	P				*				0.32	--	MMP	46
PFR IX710D75	P				*				0.34	--	MMP	46
PFR IX715DM7	P				*				0.36	--	MMP	46

Symbols

P	=	positive resist
N	=	negative resist
B	=	broadband UV exposure source
g	=	g-line UV exposure source
h	=	h-line UV exposure source
i	=	i-line UV exposure source
DUV	=	deep UV exposure source
EB	=	electron beam exposure source
XRAY	=	x-ray exposure source
C.C.	=	closed cup flash point test

Solvent Abbreviations

EEA	=	2-ethoxyethyl acetate (EGEEA) [!]
EEP	=	ethyl ethoxy propionate
EL	=	ethyl lactate
MEA	=	2-methoxyethyl acetate (EGMEA) [!]
MMP	=	methyl methoxy propionate
NMP	=	n-methyl-2-pyrrolidone
PGMEA	=	propylene glycol monomethylether acetate

[!] Solvents recommended for elimination by the Semiconductor Industry Association because of potential adverse reproductive health effects.

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Wet etching is the traditional method of etching. In this process, wafers are dipped into acid solutions or, less commonly, acids are sprayed onto the wafers. The etchants remove the surface of the wafer that is not protected by the hardened photoresist. Table 2.3 outlines typical chemistries associated with wet etching and wet stripping.

Table 2.3. Typical Wet Etch and Stripper Systems for Silicon Wafers

Etched Material		Common Etchant (and ratios)	Etch Temperature
Silicon (base wafer or polysilicon film)	Silicon Dioxide (oxide layer)	HNO ₃ - 50	Room
		H ₂ O - 20	
		HF - 3	Room
		HF - 1	
		NH ₄ F - 8	Room
		CH ₃ COOH - 2	
NH ₄ F - 1			
Silicon Nitride*		H ₃ PO ₄	170–190°C
(passivation layer)			
Aluminum		H ₃ PO ₄ - 16	40–50°C
		HNO ₃ - 1	
		CH ₃ COOH - 1	
		H ₂ O - 2	
		Wetting Agent	
Etchant Abbreviations:			
	HNO ₃	=	Nitric Acid
	HF	=	Hydrofluoric Acid
	NH ₄ F	=	Ammonium Fluoride
	CH ₃ COOH	=	Acetic Acid
	H ₃ PO ₄	=	Phosphoric Acid
* Wet etching of silicon nitride is particularly difficult, resulting in its limited use.			
Stripper Type		Etchant Mixture	Etch Temperature
Acid	Sulfuric and chromic		Elevated
		Sulfuric and ammonium persulfate	Elevated
		Sulfuric and hydrogen peroxide	Elevated
Stripper Abbreviations:			
	H ₃ SO ₄	=	Sulfuric Acid
	CrO ₃	=	Chromic Acid
	(NH ₄) ₂ S ₂ O ₈	=	Ammonium Persulfate
	H ₂ O ₂	=	Hydrogen Peroxide

Etch baths are kept in three-sided exhausted enclosures—wet benches. Typical exposures to airborne etchants at the wet benches is far below the PEL/TLVs for the respective chemicals. If a release into the fab area does occur, the odor and irritative effects of the chemicals quickly warn of the incident.

The main concern with wet etching is the possibility of skin contact with the acids. While all the acids used in etching can cause acid burns, exposure to hydrofluoric acid (HF) is of particular concern. The lag time between skin contact and pain (up to 24 hours for solutions less than 20% HF and one to eight hours for 20–50% solutions) can result in delayed treatment and more severe burns than expected.^[54] (See Ch. 5, Sec. 2.3.)

Historically, acid burns have been a particular problem within the industry. However, the incidence of skin contact with acids been reduced in recent years.^[55] Some of this reduction was caused by product-related improvements in the etch process such as the shift to dry etching and the use of more robotics. The reduction in the rate of acid burns may also be attributed to better handling techniques, greater use of personal protective equipment, better designed wet decks, and better training—all of which require continued attention if the rate is to decline further.

The requirement for smaller lines and spaces in the wafer pattern has led to a need for straighter side walls in the etched pattern (anisotropic etching). This is more easily accomplished with dry etching than wet. A wide variety of different chemistries are used in dry etching depending on the material being etched and the type of dry etcher being used (e.g., plasma etching, reactive etching, and physical etching). The majority of these chemistries are chlorine and/or fluorine based.

Because dry etchers operate as closed systems, chemical exposure to the operators of the equipment typically do not occur while the system is closed. One rare exception to this is when the purge cycle for older batch etchers is not long enough to adequately remove the etchant gases. Brief but irritating exposures to fluorine compounds that are below the detection limit for typical IH monitoring procedures have been reported when the doors to these etchers are opened. Normally this can be corrected by simply increasing the length of the purge cycle prior to opening the etch chamber door.

The major IH emphasis for dry etchers has been on potential exposures to maintenance personnel working on the reaction chambers, pumps, and other associated equipment that may contain reaction-product residues. The complexity of plasma metal etchers and the difficulty in

characterizing the odors associated with their maintenance has made them the subject of many investigations and a few published papers.^{[39][40][53][56]}

The reaction products formed in plasma metal etchers are a complex mixture of chlorinated and fluorinated compounds. The maintenance of metal etchers often involves short duration operations that generate strong odors. Hexachloroethane was found to be the major cause of odor in one type of aluminum etcher.^[39] In another, cyanogen chloride was the main problem: exposure levels were 11 times the 0.3 ppm TLV[®].^[40] In still other types of etchers, hydrogen chloride is associated with the odor (maximum exposure measured was 68 ppm; odor threshold for HCl is less than 5 ppm).^[13] The paper by Mueller and Kunes is a good source of additional information on the subject.^[56]

The complexity of the chemistries present in metal etcher exhausts has led researchers to develop experimental methods for investigating the toxicity of these mixtures.^[57] Application of these methods in inhalation studies where rats were exposed to etcher effluent six hours per day, five days per week, for four weeks showed no significant indications of organ toxicity or behavioral effect, but there were statistically significant increases in chromosomal aberrations and sister chromatid exchanges in bone marrow cells indicating these chemical mixtures are suspected mutagens.^[58]

Process: Doping. Doping involves introducing elements into the wafer to change the electrical characteristics of the silicon at locations determined by the patterns formed during lithography. Depending on the element introduced, either n-type or p-type junctions are formed in the wafer. The parent compounds of the n-type dopants are generally more toxic than their p-type counterparts. Table 2.4 outlines typical chemistries associated with doping.

Diffusion and ion implantation are the two processes used to introduce dopant elements into the wafer. In diffusion, the wafer, with the parent chemical, is heated to high temperatures (800–1300°C) in a quartz tube, and the element diffuses into the wafer. With ion implantation, a newer technology, ions derived from the parent chemical are accelerated down a beam-path and precisely implanted into the wafer.

Routine chemical exposures to operators of both diffusion furnaces and ion implanters are low—typically less than the detection limit of NIOSH sampling procedures. Chemical concerns with the process center on the possibility of toxic gas releases (see Sec. 4.2: Continuous Monitoring) and exposures to residues during maintenance activities.

Table 2.4. Characteristic Chemistries Used in Doping Silicon Wafers

Element (Type)	Typical Parent Chemicals	State
Antimony (n)	antimony trioxide (Sb_2O_3)	solid
Arsenic (n)	arsine (AsH_3)	gas
	arsenic trioxide (As_2O_3)	solid
	tertiarybutylarsine ($\text{C}_4\text{H}_{11}\text{As}$)	liquid
Phosphorous (n)	phosphine (PH_3)	gas
	phosphorous oxychloride (POCl_3)	liquid
	tertiarybutylphosphine ($\text{C}_4\text{H}_{11}\text{P}$)	liquid
	phosphorus pentoxide (P_2O_5)	solid
Boron (p)	boron trioxide (B_2O_3)	solid
	boron tribromide (BBr_3)	liquid
	boron trichloride (BCl_3)	gas
	diborane (B_2H_6)	gas
	boron nitride (BN)	solid
Gold (p)	gold (Au)	solid
Aluminum (p)	aluminum (Al)	solid

In the absence of proper controls, arsenic exposures above the 0.01 mg/m^3 exposure limit were reported during the cleaning of a deposition furnace^[9] and during the cleaning of source housing chambers for solid-source ion implanters.^[42] These exposures occurred when no precautions were taken to limit the amount of dust in the air. However, when residues were kept wet during cleaning, exposures were reduced to far below the allowable limit.^[41]

Exposures to hydrides (probably a mixture of arsine and phosphine) as high as 60 ppb have been found during ion implanter cryo-pump maintenance.^[13] Also, high concentrations of both arsine and phosphine can off-gas from contaminated implanter parts that are removed during preventative maintenance.^[59]

A novel approach for sampling and analysis of diborane gas, a fairly common hydride dopant, has been identified by absorbing concentrations as low as 0.01–3.0 ppm on silica gel impregnated with potassium permanganate as an adsorbant.^[60]

Portable vacuum cleaners with HEPA filters are used to clean arsenic-contaminated work surfaces in ion implantation areas. Exposures above 1000 $\mu\text{g}/\text{m}^3$ were measured when HEPA vacuums were improperly cleaned. HEPA vacuums, when discharging to the workspace, can also efficiently distribute the distinctive, hydride-like odor associated with ion implanter beam line cleaning.^[13]

Appendix E is a recent product support bulletin on safety considerations during maintenance following arsenic and phosphorus operations on ion implanters, and should be reviewed.

Process: Chemical Vapor Deposition. Chemical Vapor Deposition (CVD) is a process whereby the heat-induced decomposition of gases form different semiconductor layers such as silicon dioxide, silicon nitride, polysilicon, and gallium arsenide on the surface of the wafer. When the newly-formed layer is a continuation of the crystalline structure of the substrate, the process is called epitaxial growth. Other, non-epitaxial forms of CVD involve the deposition of layers that are a different structure than the substrate. Table 2.5 outlines typical chemistries associated with CVD.

In addition to the reactant gases listed in Table 2.5, hydride dopants (i.e., arsine, phosphine, diborane) may be introduced to control the type of conductivity and resistivity of the film layer. For silicon films, the dopant gases are introduced in small amounts (e.g., ppb levels) while for III–V films, the concentrations used are considerably higher (e.g., percentage concentrations). For *molecular beam epitaxy* (MBE), solid gallium is used for p-type doping and antimony for n-type doping. *Chemical vapor deposition* (CVD) units operate as a closed system resulting in little or no chemical exposure to the operators. However, brief hydrogen chloride exposure above 5 ppm can occur when certain CVD prescrubbers are cleaned.^[13]

Gallium phosphide and gallium arsenide deposition on gallium arsenide wafers require heavy concentrations of dopant gases (e.g., arsine and phosphine). Prescrubbers for these CVD reactors contain high concentrations of arsine and phosphine, and exposure far above acceptable limits can occur if these prescrubbers are indiscriminately opened.^[11]

High exposures to arsine can also occur during the maintenance of solid-source molecular beam epitaxy (MBE) systems for gallium arsenide. Room air concentrations of 0.08 ppm were detected in one study when the chamber of the MBE unit was opened for maintenance. The authors hypothesized that transient arsine generation may be caused by a reaction of very fine particulate arsenic with water vapor, with aluminum acting as a catalyst.^[38]

While Group III–V (e.g., gallium phosphide and gallium arsenide) and Group II–VI (e.g., mercury cadmium tellurium) semiconductor devices are far less common than silicon semiconductors, CVD operations for them have certain chemistries unique in semiconductor manufacturing.^{[36][61]–[63]} Halide CVD processes used in the manufacture of some III–V devices utilize arsenic trichloride and phosphorus trichloride as source chemicals.^{[61][62]}

Table 2.5. Typical CVD Chemistries

CVD Type	Film	Reactant (Carrier Gas)
Atmospheric Pressure (APCVD)	Epitaxy Si	SiCl ₄ /H ₂ (H ₂) SiHCl ₃ (H ₂) SiH ₂ Cl ₂ (H ₂) SiH ₄ (H ₂)
	Poly Si	SiH ₄ (H ₂)
	Silicon Dioxide	SiH ₄ /O ₂ (N ₂)
	Silicon Nitride	SiH ₄ /NH ₃ (H ₂) SiH ₂ Cl ₂ /NH ₃ (N ₂)
Low-Pressure (LPCVD)	Epitaxy Si	SiH ₂ Cl ₂ (H ₂) SiH ₄ (H ₂)
	Poly Si	SiH ₄ (H ₂)
	Silicon Dioxide	SiH ₂ Cl ₂ /N ₂ O SiH ₄ /N ₂ O SiH ₄ /O ₂
	Silicon Nitride	SiH ₄ /NH ₃ (N ₂) SiH ₂ Cl ₂ /NH ₃ (N ₂)
Low-Pressure Plasma Enhanced	Silicon Nitride	SiH ₄ /NH ₃ (N ₂)
Molecular Beam Epitaxy (MBE)	Gallium Arsenide	Ga/AsH ₃
Vapor Phase Epitaxy (VPE)	Gallium Arsenide	AsCl ₃ /H ₂ /Ga
Metalorganic (MOCVD)	Gallium Arsenide	(CH ₃) ₃ Ga/AsH ₃

CVD Abbreviations: SiCl₄ = Silicon Tetrachloride; H₂ = Hydrogen; SiHCl₃ = Trichlorosilane; SiH₂Cl₂ = Dichlorosilane; SiH₄ = Silane; O₂ = Oxygen; N₂ = Nitrogen; N₂O = Nitrous Oxide; NH₃ = Ammonia; AsH₃ = Arsine; Ga = Gallium; AsCl₃ = Arsenic Trichloride; (CH₃)₃ Ga = Trimethyl Gallium

Metallorganic chemical vapor deposition (MOCVD) is now widely used in the preparation of III-V and II-VI devices. In addition to the hydride gases used as a source materials in other CVD systems (e.g., arsine and phosphine), less toxic liquid alternatives (e.g., tertiary butyl arsine and tertiary butyl phosphine) are also used in MOCVD systems along with other toxics such as cadmium alkyls and mercury.^{[61][62]}

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Process: Metallization. Metallization is a deposition step that connects separate parts of the device to perform circuit functions such as surface wiring. Vacuum systems such as evaporation and sputtering are used to deposit the metal layers. Some metal layers used in metallization are:

Aluminum	Tungsten
Aluminum-silicon alloys	Thallium
Aluminum-copper alloy	Molybdenum
Aluminum-titanium	Gold
Titanium-tungsten	Platinum
Titanium	Silver

Removing the metal residues that deposit on the inside surface of evaporator deposition chambers (bell jars) may result in significant exposures. For example, technician exposures far above the TLV for silver were measured during residue removal from an evaporator used to deposit silver.^[11]

Cleaning bell jar residue with organic cleaning solvents can result in high solvent exposure. Technician exposures to methanol above 250 ppm have occurred during this type of cleaning. This exposure can be eliminated by using water as the cleaning solvent instead of methanol.^[11]

Process: Assembly and Test. Once fabrication is complete, each IC on the wafer is powered up and tested. Sub-standard ICs are marked with a drop of ink deposited by the test device. The wafer is then cut apart and separated into individual chips ready for assembly.

The chips are mounted in ceramic or metal packages, or on lead frames that are encapsulated in a plastic molding compound. The plastic epoxy molding compounds used in encapsulating the IC chip are based on epoxy Novolak[®] resins with antimony trioxide (1–5%) added as a fire retardant. Particulate exposures are of concern to molding press operators, and to QC/QA technicians performing chemical analysis testing. Ceramic packages may contain beryllium as an alloying agent.^[64] When packaged, the IC is again tested in a series of computer-generated tests. Samples are also pulled for reliability testing. If the samples fail the tests, failure-analysis is performed to discover the root cause of the problem.

These final steps in the manufacturing process involve the application of many different operations that include: soldering, degreasing, testing with chemicals and radiation sources, and trimming and marking with lasers.

Soldering during semiconductor manufacturing normally does not result in high lead exposures. To prevent thermal damage to the integrated circuit, the solder temperature is kept below the temperature where significant molten lead fuming can occur (430°C). However, cleaning solder equipment by scraping or brushing the lead-containing residues can result in lead exposures above 50 µg/m³.^[11] Also, lead exposures of 200 µg/m³ have occurred when improper dross removal techniques are used during wave solder operations.^[65]

One growing concern with solder operations is exposure to the pyrolysis products of the solder fluxes, particularly during hand soldering or touch-up operations where historically local exhaust ventilation has not been commonly used (unlike wavesolder operations that for the last few decades have typically been enclosed in exhausted cabinets). The basic constituent of most solder fluxes is colophony (rosin). Pyrolytic decomposition of the rosin results in the formation of oxidation products of abietic acid and possibly dehydroabietic acid, and levopimaric acid. The general term of colophony fume is used to describe this chemical mixture. This material is irritating and has been found to be a contact allergen, possibly by sensitization through insertion into the lipid bilayer.^{[66]–[70]}

The *ACGIH TLV Booklet* lists “rosin core solder thermal decomposition products as resin acids—colophony” as a “sensitizer; reduce exposure to as low as possible.” The recent British “Approved Code of Practice, Control of Respiratory Sensitisers,” also considers solder flux a respiratory sensitizer (along with spores from moldy hay, isocyanates, flour dust...),^[71]

Since colophony is a sensitizer, all exposures should be reduced to as low as possible regardless of air sampling results. New soldering installations particularly should include local exhaust ventilation when soldering is to be performed for extended periods of time (e.g., >2 hours).

Fumes from hand soldering will rise vertically on thermal currents entering the employees breathing zone as the person leans over the point of soldering. Control is normally achieved by means of effective high velocity and low volume local exhaust ventilation at the solder tip.

Devices that return filtered air to the workplace may, if the filtration efficiency is inadequate, cause secondary pollution which can effect people in the work room other than those soldering. Filtered air therefore should not be returned to the work room unless the amount of soldering is small and the room has good general dilution ventilation.

Various types of exhaust ventilation systems can be used to control soldering fumes. A short review of those systems are as follows:

Exhaust Nozzles Fitted to the Soldering Iron. This system involves the mounting of a small nozzle on the soldering iron parallel to the soldering tip. The system is high velocity and low volume in its design. The advantage is that the solder fume is captured at the point of origin and before it can enter the breathing zone of the employee. The disadvantage of using such a system is that the fume residue blocks the system suction tubes and filters on a regular basis requiring ongoing maintenance.

Exhaust Systems Using Capture Hoods. These are conventional ducted exhaust ventilation systems comprising central collection ducts with flexible branch ducts running to the capture hoods at each soldering position. The effectiveness of these systems is directly related to how close the hood is placed to the immediate work area. Control can be achieved with relatively low capture velocities of 50 fpm, but the hood must be positioned within 4 to 6 inches (100–150 mm) of the fume source. Unless the hood is self-supporting, it may be difficult to accommodate large work pieces. Other problems that should be considered include the noise level produced by the fan blower unit and heat loss to the building.

Local exhaust ventilation should also be provided at wave solder machines and at any area where soldered boards cool and still evolve colophony fume.

Individual Fume Control Units. These units are often called fume absorbers and are readily available from a number of suppliers. They are normally comprised of a box housing a fan and filtering device. The fans are generally small and low-powered making it necessary that the unit be placed close to the work piece, about 4 to 8 inches (100–200 mm). Claims for efficiency of these units should be regarded as suspect. In addition, these units require on-going maintenance involving filter replacement.

Although these units are not regarded as a top choice in the elimination of contaminants from the workplace they may be acceptable in some cases where soldering is limited, the

general room ventilation is adequate, and the unit can be placed close to the work piece.

Exhaust-Ventilated Benches. A high standard of fume control can be achieved by the use of ventilated work benches. Fume control is possible at capture velocities as low as 50 fpm (0.25 m/s), but in practice, the system is best designed to provide a capture velocity of 100 fpm (0.5 m/s). The main drawbacks of using exhaust-ventilated benches are cost and the design of the room layout, particularly when installing benches in an existing workroom.

Degreasers have been used in the past to clean semiconductors before they are marked with identification codes. Solvent exposure above the applicable PEL/TLV can easily occur if an operator's head is placed below the cooling coils that cause the vapors to recondense. This exposure can happen when an operator attempts to retrieve dropped parts or when a technician cleans residue from the bottom of the unit.^[11] The use of degreasers has been greatly reduced in the semiconductor industry due to restrictions on the use of ozone-depleting substances such as chlorofluorocarbons and chlorinated solvents.

QA and reliability labs typically perform various operations that present the potential for employee exposure. Some examples are:

- Marking tests—utilizing various solvent and corrosive mixtures in heated beakers on hot plates. The provisions for local exhaust ventilation (LEV) in the form of a metal hood with adequate face velocities are needed to control fugitive emissions. Monoethanolamine solutions can result in exposures in excess of the PEL/TLV.^[45]
- Bubble/leak testing—utilizing high molecular weight fluorocarbons (trade name Fluorinerts®). This usage is being phased out due to ozone depletion concerns.
- X-ray packaging units (see Ch. 3.0: Physical Agents, Ionizing Radiation, x-ray generating equipment section).
- Fine leak detection - using krypton⁸⁵ radioactive gas—a beta and gamma emitter (see Ch. 3.0: Physical Agents, Ionizing Radiation, radioactive materials section).

Finally, laser package markers which have Class 4 carbon dioxide lasers (Class 1 with the equipment enclosure in-place and the door(s)

closed) are used to physically mark the exterior of plastic and ceramic packages. Fumes can be created during the marking and create residues on the equipment surfaces and fume extraction filters. Plastic epoxy packages contain antimony trioxide as a fire retardant (typically 1–5%), and ceramic packages may contain beryllium.

Process: Cluster Tools. Many cleanroom processes can be combined into what is termed a *cluster tool*. These are discrete functional modules linked to a central robot that allows the semiconductor device manufacturer to purchase one process tool that may perform dry etching, metallization and passivation in one piece of equipment. Figure 2.3 provides a diagram of one equipment manufacturer's cluster tool incorporating CVD, annealing, poly cleaning and cooling.^[72] By 1995, cluster tools are expected to make-up greater than 60% of the semiconductor thin-film processing equipment market.^[72]

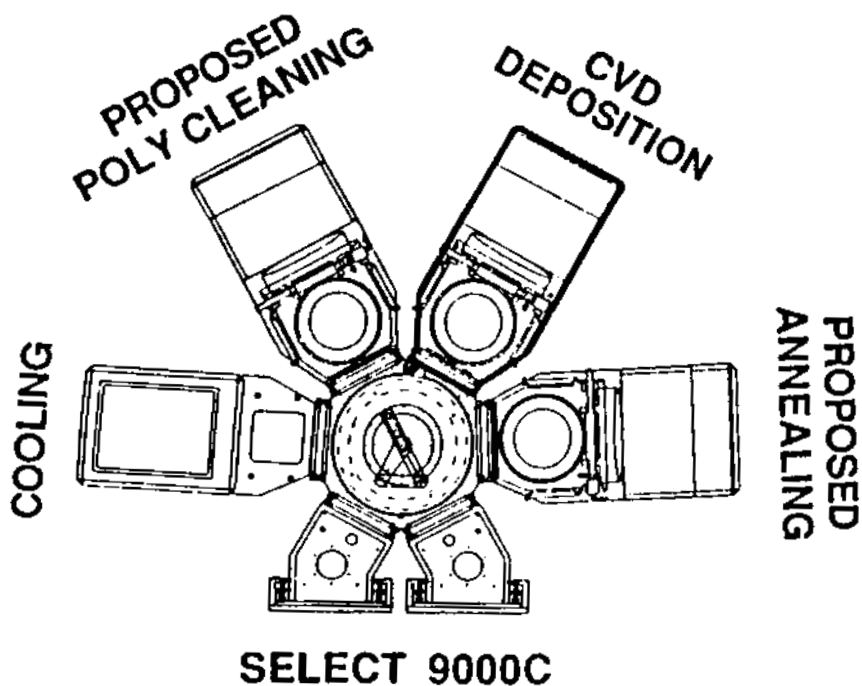


Figure 2.3. Cluster tool schematic layout from equipment vendor.^[72] (© 1993, ACGIH, reprinted with permission.)

From an IH perspective, cluster tools tend to have concerns comparable to the processes occurring on standard fabrication equipment. One exception to this is the possibility of accidental mixing of incompatible chemicals from separate modules that share a common chamber.^[72] This concern is being addressed in the design of the equipment. The other concern relates to the exhaust effluents that may be going up a single treated exhaust system. Exhaust conditioning equipment should be investigated closely to ensure that it can handle the multiple chemical species that may be formed, and that it is designed for ease of maintenance and safety.

Process: Vacuum Pumps. Vacuum pump systems are used throughout semiconductor processing. Some examples are: plasma etching and ashing, ion implantation, low pressure CVD, and metallization. The major categories of vacuum pumps are categorized in Table 2.6. The historical usage sequence in the semiconductor industry has been: diffusion, ion getter pumps, turbopumps and cryopumps with the emergence of the turbopump as the major pump choice of IC manufacturing.^[73]

Mechanical type oil-seal rotary roughing pumps in conjunction with either in-line oil diffusion, or cryogenic pumps are used. The use of independent turbomolecular pumps has been increasing. There is also a tendency in some processes to use dry pumps (i.e., those that do not require pump oils) in place of pumping processes that had previously required the use of expensive inert perfluoropolyether fluids,^[74] which had been a replacement for the still older technology of hydrocarbon oils used in conjunction with a cold trap (a cryogenic method of removing reactive gases prior to contact with the pump).

IH concerns associated with vacuum pumps center on off-gassing during pump oil changes and during regeneration of cryo-pumps. This off-gassing can be the parent compounds used in the process, reaction products formed as part of the production operations, reaction products formed between the gases being pumped and the pump oil, or a mixture of these categories. As previously mentioned in the doping section, significant airborne hydride levels have been found during ion implanter cryo-pump maintenance when regeneration was done in other than a closed system.^[13]

Although there is a concern, there have been no published reports of significant dopant gas exposures during oil changes of vacuum pumps used with dopants—possibly because this is usually done as a closed system. The lack of reported exposure may also be a result of low levels of off-gassing of hydrides from the used oil.

Table 2.6. Vacuum Pump Types

a. Mechanical pumps
1. Oil-seal rotary pump (roughing)
2. Roots-pump
3. Molecular-drag pump
b. Oil diffusion pumps
c. Turbomolecular pump
d. Cryogenic pumps
e. Crysorption pumps
f. Getter pumps
1. Ion pump
2. Sublimation pump
3. Evaporation pump
4. Sputter-ion pump

Figure 2.4 shows the result of a field study where 700 ml of used roughing pump oil from an ion implanter which used both arsine and phosphine was heated.^[42] Head space analysis of the heated oil only showed detectable concentrations of airborne hydrides when the pump oil exceeded 70°C. Since normal operating temperatures for mechanical roughing pumps are 60–80°C, this study did not indicate the potential for a significant exposure.

4.2 Continuous Gas Monitoring

A large number of toxic and hazardous gases are used in semiconductor manufacturing operations around the world. Materials like arsine, diborane, phosphorous pentafluoride, phosphine, chlorine, fluorine, hydrogen chloride, and silane are essential to a variety of electronic device fabrication processes. Typical processes utilizing these materials include chemical vapor deposition, ion implantation, reactive ion etching, and sometimes even photolithography. These gaseous materials are not limited to only silicon-based IC production facilities. They can also be found in private and public sector research and development facilities investigating a broad range of advanced electronic device technologies such as optoelec-

tronics, vacuum electronics, flat panel displays, and microwave/millimeter-wave solid state devices. Since these gases are hazardous to both human beings and the environment, stringent regulations have been developed (and will continue to be developed) which govern their transportation, storage, use, and disposal.

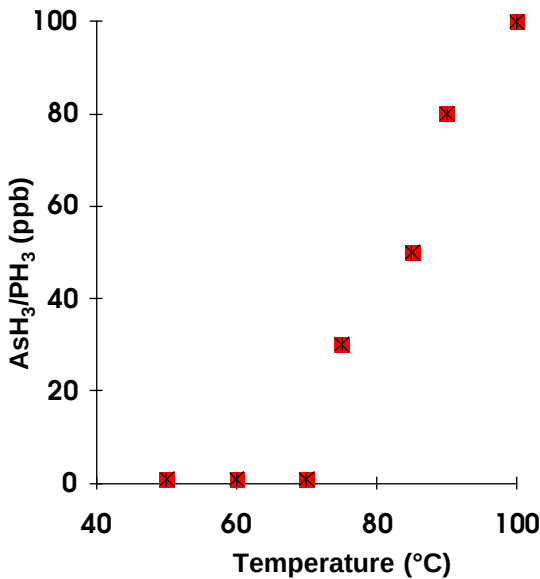


Figure 2.4. Headspace analysis of heated roughing pump oil.^[42]

The use of continuous monitoring for determining compliance with governmental personal sampling requirements is an area of conjecture. Recent studies have shown that, under certain circumstances, continuous monitoring systems can substitute for compliance monitoring. One study at a petrochemical facility showed good correlation between results of personal sampling and continuous monitoring system ($r = 0.70$ and 0.92).^[75] Another study done in conjunction with the recent Johns Hopkins epidemiological and IH study of IBM's U.S. Semiconductor Operations showed ICAMTM mass spectrometer monitoring overestimated personal sampling results and therefore could be used to demonstrate compliance with governmental exposure regulations.^[76]

While most industrial hygiene air monitors in the semiconductor industry are used for flammable and toxic gas leak detection, some facilities are also using continuous monitoring systems to:

- Analyze exhaust duct (stack) emissions
- Quantify ambient air concentrations of volatile chemicals
- Identify and quantify odors in the fab areas (see Ch. 6: Indoor Air Quality, Sec. 6.2: Odor Identification, for additional discussion)

The technologies most used in the semiconductor industry for this type of monitoring are mass spectrometry (e.g., Perkin-Elmer ICAMS™) and fourier transform infrared (e.g., Telos FTIR).^{[77]–[81]} The detection limits for ICAM and FTIR systems are similar. For compounds used in semiconductor manufacturing, the detection limits needed are typically less than 0.1 ppm for single airborne volatile organic compounds. The detection limit is often one or two orders of magnitude higher for mixtures, inorganics, and when sampling for unknowns.^{[79][80]} The other major limitation of continuous monitoring—aside from cost—is the inability to detect certain compounds. Some facilities approach this by purchasing multiple types of monitors (e.g., one for hydrides: arsine, phosphine, diborane, and silane, one for organics, and compound specific monitors for other chemicals such as nitrogen trifluoride). This is an effective but costly solution.

Brief History. As early as the 1970s, progressive semiconductor manufacturers began installing the first continuous gas monitoring systems for flammable and toxic gases. The main focus of this monitoring was to detect accidental releases of the most toxic gases with odor thresholds above the TLV concentration (e.g., arsine and diborane).

The installation of continuous gas monitoring systems by semiconductor manufacturers predates specific code requirements. In fact, the codes adopted in the 1980s were based on the practices developed by the semiconductor industry and equipment manufacturers. The majority of these early code revisions and additions took their roots from actions initiated by local fire and safety officials in “Silicon Valley.” In 1982, the City of San Jose began enforcing fire and building codes which addressed specific aspects of semiconductor wafer fabrication facilities. The City of Santa Clara passed a Model Ordinance, commonly known as the “Green Book,” specifying building codes for semiconductor operations.^[82] Larger regional code administrators, including the International Conference of Building Officials (ICBO), the Western Fire Chiefs Association (WFC),

the Building Officials and Code Administrators International (BOCA), and the Southern Building Code Congress International (SBCCI), began developing fire and building code requirements for semiconductor wafer fabrication facilities using hazardous production materials. Code modifications resulting from these local and regional efforts included Article 51 and Article 80 of the Uniform Fire Code (UFC), Sec. 603 of the BOCA National Building Code, Ch. 9 of the Uniform Building Code (UBC), and the Toxic Gas Model Ordinance.

The Toxic Gas Model Ordinance was written by the Santa Clara County Fire Chief's Association to serve as an accompaniment to UFC Article 80. It required that "All new or existing storage, use or handling of toxic gases shall ...provide a monitoring and detection system. Such systems shall be capable of sensing the presence of toxic gases at or below TLV-TWA levels. Audible alarms shall be located within the local storage and use areas and activate when the concentration of toxic gas exceeds the TLV-TWA level in occupied areas or one-half the IDLH level in unoccupied areas. Alarms shall transmit to an emergency control station." It defined a monitoring system as a continuously operating gas detection system where an analytical instrument is maintained in operation and samples are taken at appropriate points and analyzed without interruption for the presence of toxic gases.^[83]

Similarly, the Revisions to UFC Article 80 required that "a continuous gas-detection system shall be provided to detect the presence of gas at or below the permissible exposure limit or ceiling limit. The detection system shall initiate a local alarm and transmit a signal to a constantly attended control station. The alarm shall be both visual and audible and shall be designed to provide warning both inside and outside of the storage area. The audible alarm shall be distinct from all other alarms." It defined a continuous gas detection system as "a gas-detection system where the analytical instrument is maintained in continuous operation and sampling is performed without interruption. Analysis may be performed on a cyclical basis at a frequency not to exceed 30 minutes."^[84] (It should be noted that most systems used in the semiconductor manufacturing have cycle times that are significantly less than 30 minutes).

While various codes may differ in definition and/or scope, there is a recurring requirement for continuous gas detection by a *toxic gas monitoring system* (TGMS).

Purpose of a Toxic Gas Monitoring System. The primary purpose of a gas detection system is to continuously monitor an area for the

presence of hazardous and toxic gases at or below a specific concentration limit, and if necessary, warn personnel through both visual and audible alarms prior to the onset of a life threatening condition and in some cases to automatically shut-off the gas supply.^[85]

There are several key terms identified in this single summary statement that must be understood.

Continuous monitoring means uninterrupted monitoring, occurring in real-time.

Area refers to any physical space or location where there is a potential for a gas leak. It includes both “occupied” areas, like laboratories routinely used by researchers or cleanroom fabs containing process operators, and “unoccupied” areas, like gas supply rooms which are infrequently visited by support personnel. It can be a confined space, such as the inside a gas cabinet or furnace end station, or an open area, such as the work space around an ion implanter or service area which contains diffusion pumps and exhaust lines for a chemical vapor deposition (CVD) system.^[86]

Specific concentration limit refers to a critical gas concentration limit that should not be exceeded. This limit can be a Threshold Limit Value (TLV), Time Weighted Average (TWA) or Ceiling (C); Permissible Exposure Limit (PEL); TLV, Short Term Exposure Limit (STEL); Lower Explosive Limit (LEL) or Immediate Danger to Life and Health (IDLH) limit. These limits can be defined in terms of percent volume (%), parts per million (ppm), or even parts per billion (ppb). Table 2.8 lists hazards and exposure limits for several common hazardous gases found in the electronics industry. Critical gas concentration limits may be mandated by environmental, health and safety statutes,^[87] regional or local codes, local code enforcement officials, internal risk management personnel, or an established organizational policy.

Personnel refers to anyone who could be exposed to hazardous or toxic gases if a gas leak occurs. These people might include researchers working in a laboratory, process engineers and technicians working in a cleanroom fab, equipment specialists performing corrective or preventative maintenance, facility support personnel changing a gas cylinder, emergency

Table 2.7. Hazards and Exposure Limits for Common Gases Used in Electronic Device Processing.

Gas	Chemical Formula	Hazard Classification*	TLV-TWA, C or STEL in ppm or other caution from ACGIH**
Ammonia	NH ₃	F,C	25 (35 STEL)
Argon	Ar	I	Asphyxiant
Arsenic Pentafluoride	AsF ₅	T	TLV not listed (use HF)
Arsine	AsH ₃	T,F	0.05
Boron Trichloride	BCl ₃	T,C	TLV not listed (use HCl)
Boron Trifluoride	BF ₃	T,C	TLV not listed (use HF)
Chlorine	Cl ₂	T,C	0.5 (1 STEL)
Diborane	B ₂ H ₆	T,F,P	0.1
Dichlorosilane	H ₂ SiCl ₂	F,C	TLV not listed (use HCl)
Fluorine	F ₂	T,C,F	1
Germane (Germanium Tetrahydride)	GeH ₄	T,F	0.2
Hydrogen	H ₂	F	Asphyxiant (monitor as a flammable gas)
Hydrogen Chloride	HCl	C	5C
Hydrogen Fluoride	HF	T,C	3C
Hydrogen Sulfide	H ₂ S	T,F	10 (15 STEL)
Methyl Chloride	CH ₃ Cl	F	50 (100 STEL)
Nitric Oxide	NO	T,O	25
Nitrous Oxide	N ₂ O	O	50
Nitrogen Trifluoride	NF ₃	T	10
Phosphine	PH ₃	T,F,P	0.3 (1 STEL)
Phosphorous Pentafluoride	PF ₅	T,C	TLV not listed (use HF)
Silane	SiH ₄	P,F	5
Sulfur Hexafluoride	SF ₆	I	1000

Notes:

* From Semi-Gas Systems Gas Reference Guide 1986: F=flammable, C=corrosive, I=inert, T=toxic, P=poison and O=oxidizer (with updating)

** From Threshold Limit Values and Biological Exposure Indices for 1993–1994, American Conference of Governmental Industrial Hygienists (ACGIH). (In this publication the ACGIH issues strict caution on the use of TLV values, for example, they should not be used by anyone untrained in the discipline of industrial hygiene.)

response personnel such as HAZMAT team members and security guards, or any other personnel who might be affected in the immediate or near-by, adjacent areas.

Visual and audible alarms refers to whatever type of appropriate human sensory inputs are required in a specific area to gain the attention of personnel and inform them that a hazardous condition exists. This usually means alarms in the form of bright flashing or highly visible strobe lights, distinctive sounding horns, buzzers, or sirens, or combination of any of the above. In cleanroom areas, where air velocity noise and full hood or helmet garment protocol can make hearing difficult, visual alarms may provide the first warning. Similarly, in support areas where equipment specialist may be underneath or inside enclosures while performing maintenance, audible alarms may provide the first warning.

In addition, the gas detection system should also provide meaningful annunciation and appropriate record keeping. An alarm situation should be instantly recognizable as to its type, location and severity. Contaminated areas should be accurately identified, their entrances and exits clearly marked, and safe exit paths designated. Trend information, malfunction data, and service history should be readily available for analysis so that appropriate response action can be taken.^[88] The gas detection system can also initiate automatic shut-off of the gas source near the CGA connection.

Functional Model of a Toxic Gas Monitoring System. A basic gas detection system can be broken down into several distinct functional components as shown in Fig. 2.5:

1. Transport of Air Sample
2. Analysis of Air Sample
3. Identification of Target Gas Concentration
4. Comparison of Concentration to Alarm Setpoints
5. Actions to Alarm Conditions.

Transport. The transport functional component represents how an air sample from the monitored area atmosphere is transported to the gas detection analyzer. The majority of present day gas detectors use a sample draw-line technique.^[89] An air pump, usually integrated in the gas detector system package, draws a sample of air through an open-ended tube, located in the monitored area, to the analysis instrument. The tubing can be

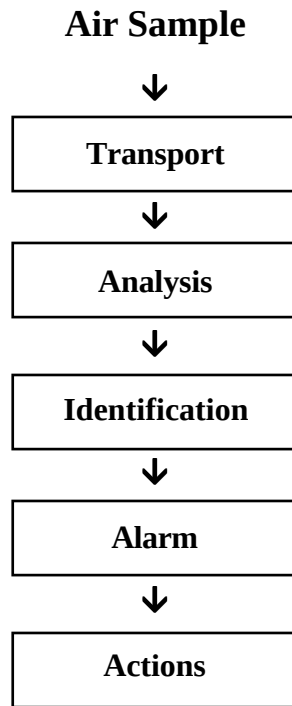


Figure 2.5. Five functional components of a basic gas detection system.

a variety of diameters and lengths. The diameter of the tubing must be compatible with the gas detector intake port and air pump. Larger tube diameters reduce flow restriction and air pump loading. Typical sample draw-lines have internal diameters of 3/16 inch to 2 inches. The maximum allowable length of a sample draw-line is dependent on tube diameter, air flow restriction, and air pump capacity. Single-point gas detectors, located in the immediate monitored area, may have a single sample draw-line with a length of less than a few inches. Remotely located multi-point gas detectors may have several sample draw-lines, each with a length of up to 1000 feet.^[90] An important point to remember is that a long sample draw-line increases the overall gas detection system response time. Also, the target gas (i.e., the gas of interest being monitored) should not react with the sample draw-line tubing material. Some gas detectors rely on other transport techniques, different than a sample draw system, to deliver the air

sample to the analysis instrument. For example, a single-point electrochemical diffusion sensor, located in the immediate monitored area, might use a simple chemical isolation diffusion barrier. Monitored gases diffuse through the gas-specific membrane and enter the electrochemical cell for analysis. Whatever method is used, once an air sample is transported to the gas detector, it must be analyzed for the presence of a target gas.

Analysis. The analysis functional component represents the gas detection principle or technique used by the gas analyzer to determine if a specific target gas is present in the air sample. Many different types of gas detection techniques can be employed to identify a target gas, but all gas detection techniques may not be appropriate for detecting certain target gas species or specific low-level target gas concentrations. Some gas detection techniques can analyze only a single gas while other technologies can analyze several gases simultaneously. Gas detectors, both single and multi-point, can be further categorized into these two additional sub-groups which are sometimes referred to as single gas analyzers and multi-gas analyzers, respectively.

Gas Detection Principles. Some of the more common gas detection principles used in toxic gas monitoring include paper tape, electrochemistry, electro-optical detection, solid state detection, mass spectrometry, molecular (or flame) emission spectrometry, infrared spectrophotometry, ionization techniques, thermal conductivity, catalytic, and chemiluminescence. Brief descriptions of several different gas detection technologies are described below.

Electrochemical Cell. As shown in Fig. 2.6, an electrochemical cell, consisting of two electrodes immersed in a common electrolytic medium, is separated from the outside world by an isolation barrier. Typical electrolytic mediums include an electrolytic liquid, an electrolytic fluid of gel-like consistency, or a porous solid which is impregnated and saturated with an electrolytic solution. Typical isolation barriers include a gas-specific permeable membrane, a diffusion medium, or a capillary diffusion barrier. A voltage is impressed across the two electrodes in the cell. A target gas enters the cell through the isolation barrier and causes an oxidation-reduction reaction to occur on the electrodes. The result is a small electrical current proportional to the partial pressure of the target gas in the ambient air. The electrochemical cell, isolation barrier, electrolytic medium, and electrodes are designed for maximum sensitivity to the target gas with minimum interference from other gases or chemical vapors which might be present.

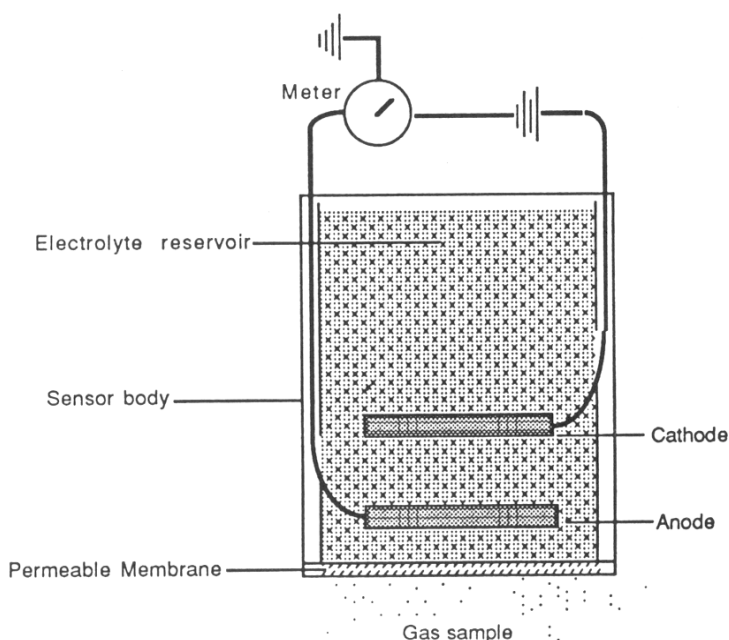


Figure 2.6. Enlarged cross-sectional view of an electrochemical sensor. (Reprinted with permission from the Semiconductor Safety Association.)

Paper Tape. A chemically impregnated detection tape is continually fed past and exposed to air samples from a sample draw-line as shown in Fig. 2.7. The tape is specially formulated to chemically react only in the presence of a target gas. When the tape is exposed to a target gas, the tape changes color in direct proportion to the amount of gas present. This color change becomes increasingly dark for higher concentrations of a target gas. An electro-optical detection system continually measures this color change by the amount of light reflected off the tape. The photo-response is proportional to the partial pressure of the target gas in the ambient air.

Solid-State. A solid-state sensor consists of two electrodes imbedded into a metal oxide material as shown in Fig. 2.8. Operating at a constant power, the sensor has a fixed resistance between the two electrodes. When the target gas is present, it causes a change in resistance of the solid state material proportional to the target gas concentration. The effects of interferent gases and chemical vapors are minimized through the selection of the metal-oxide sensor material and sensor fabrication techniques. A heating source is used to maintain a constant surface temperature on the metal oxide material and eliminate resistance changes due to variations in external temperatures.

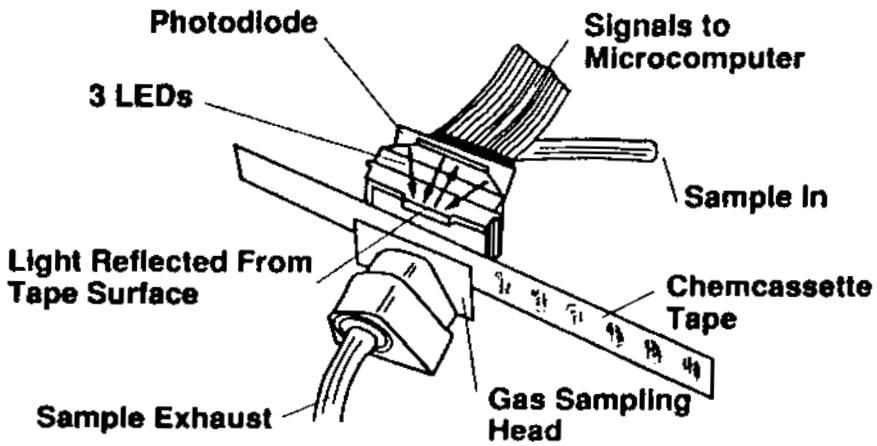


Figure 2.7. Paper-tape detector head schematic. (Reprinted with permission from the Semiconductor Safety Association and MDA.)

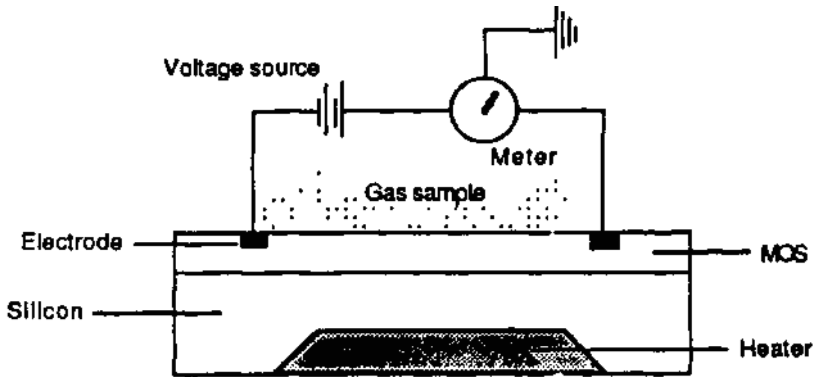


Figure 2.8. Cross-sectional and enlarged view of a metallic oxide semiconductor (MOS) solid state detector in a silicon wafer. (Reprinted with permission from the Semiconductor Safety Association.)

Mass Spectrometry. As shown in Fig. 2.9, mass spectrometry detectors use a controlled sample draw-line system in conjunction with a mass spectrometer analyzer. A specific air sample volume enters the system's ionization chamber. The sample molecules are bombarded by electrons, and fragmented into ions. These ions are transported in a beam by electrostatic and magnetic fields and focused in both energy and angle spread. The ions are dispersed according to their mass-to-charge ratio, and directed to either a Faraday cup collector or high sensitivity electron multiplier detector. The resulting intensities at the collector and detector directly correspond to the compound concentrations being analyzed and mass spectra can be used to determine compound identities. Knowledge of all compounds routinely present in the monitored atmosphere is essential to avoid false readings from interferents.

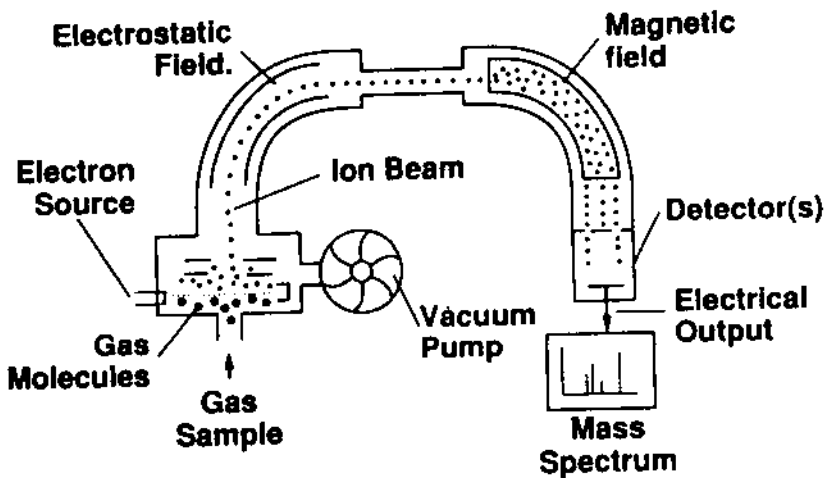


Figure 2.9. Diagram of a mass spectrometer showing ionization of gas sample molecules. (Reprinted with permission from the Semiconductor Safety Association and Perkin-Elmer.)

Molecular Emission Spectrometry (or Flame Emission Spectrometry). As shown in Fig. 2.10, an air sample from a sample draw-line is introduced into a reaction chamber, where it is mixed with hydrogen gas and activated through combustion. Chamber conditions are set to raise the sample gas molecules to excited energy states and give off light of specific

wavelengths. These wavelengths are characteristic of excited gas molecule compounds. All wavelengths of light, except the specific target gas wavelengths, are eliminated using pre-determined optical filters. A photomultiplier tube interprets the intensity of the target gas radiation which is proportional to the compound concentration being analyzed. The choice of specific wavelength optical filters can minimize the effects from interferent gases.

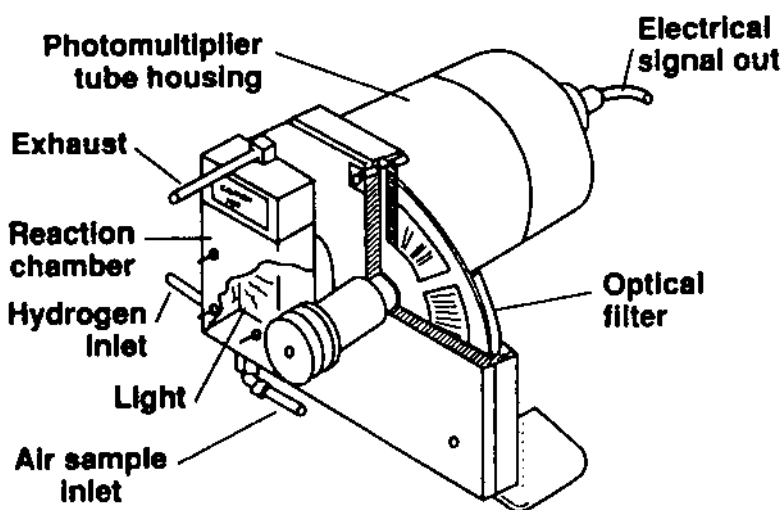


Figure 2.10. Molecular (or flame) emission spectrometer instrumentation sensor. (Reprinted with permission from Solid State Technology.)

Infrared (IR) Spectrophotometer. As shown in Fig. 12.11, Spectrophotometric IR sensors introduce an air sample into the gas cell of an infrared spectrometer. Inside the gas cell is a system of lenses and mirrors that directs a beam of monochromatic infrared light in a preselected path through the sample. The amount of energy absorbed by the sample is measured. In addition, the same sample is examined at other wavelengths. This multi-wavelength, multicomponent absorbance data is analyzed by a built-in microprocessor to determine the concentration of the compound being detected.

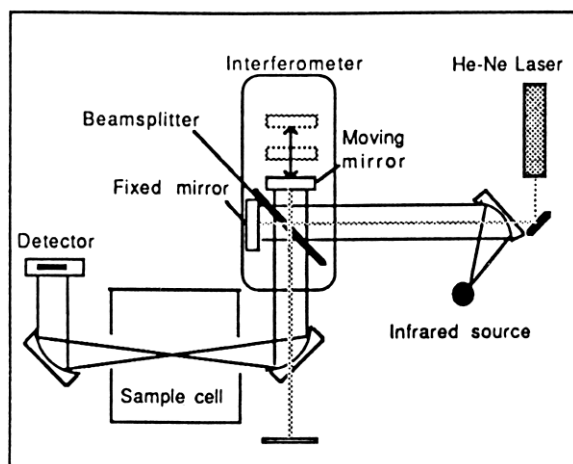


Figure 2.11. Diagram of a fourier transform infrared spectrometer (FTIR). (Reprinted with permission from the Semiconductor Safety Association.)

Flame Ionization. An air sample from a sample draw-line passes through a hydrogen flame within an ionization cell. Any organic molecules that pass through the hydrogen flame produce ions. These charged particles immediately pass through a electric field creating a current flow which is proportional to the total hydrocarbon concentration in the air sample.^[91]

Catalytic Reaction. A bead or wire is coated with a catalytic material so that it reacts with a target gas. As the reaction on the catalyzed surface takes place, the bead or wire heats up, and changes its resistance. This resistance change can be proportionally related to the target gas concentration. An example of a catalytic bead detector is a sensor that consists of two beads placed in a Wheatstone bridge circuit. One of the beads acts as a reference and the other bead is catalyzed so that it oxidizes combustible gases. As the sensing bead oxidizes, it heats up, and changes its resistance relative to the reference bead. This resistance change causes a change in the output voltage signal which is proportional to the combustible gas concentration.

Chemiluminescence. Chemiluminescence detectors are similar to infrared analyzers, but instead of measuring the wave absorbed by the target gas, it measures light emission from the gas. The magnitude of this light emission is proportional to the gas concentration.^[92]

Identification. The identification of functional components represents the method by which the concentration of an identified target gas is determined. This determination is usually made by comparing the gas detector's response to the target gas to an already known response of a specific calibration standard. Using the paper tape gas detector method described earlier as an example, if a target gas causes a measurable dark stain to appear on the chemically impregnated section of detection tape, the photo-response of this stain can be compared to the known photo-response of a specific calibration standard. Through this comparison, the concentration of the target gas can be determined.

Accuracy. The accuracy of this concentration measurement is dependent on:

1. The merits of the known calibration standard
2. The repeatability of the detection measurement technique to identical target gas concentrations
3. The complete understanding of response and precision across the entire range of the gas detection measurement instrument
4. The distortion of the measurement caused by interferences

If the calibration standard is affected by external factors (for example, variations in output from a gas diffusion tube caused by temperature changes), it may not have the required or assumed concentration value when calibrating the actual gas detection instrument. The measured target gas concentration must be questioned if the detection technique does not produce repeatable results. In the paper tape gas detector example, the accuracy and repeatability of this measurement is directly dependent on the chemical characteristics and uniformity of the impregnated detection tape, and the calibration of the electro-optical detector. Knowing the output of a gas detection instrument at a zero reading level (i.e., when no target gas is present) and the output at a maximum scale level (using a known standard having the maximum scale concentration) does not imply that an output halfway between the zero and maximum output levels corresponds to a target gas concentration of half the maximum scale value. This is only true if the gas detector's response output is linear between the zero reading level and the maximum scale level. If the gas detector's response output is not linear, the target gas concentration reading must reflect the non-linearity of the response.

Example: A gas detector has:

Zero Output Level = 4 mA

Maximum Scale Output Level = 20 mA

Maximum Scale Value = 10 ppm (parts per million)

The output level for a target gas concentration of 5 ppm would be 12 mA only if the gas detector's response output was linear as shown by the solid line in Fig. 2.12.

If the gas detector's response output was not linear and had the response output characteristic as shown by the dotted line in Fig. 2.12, the output level for a target gas concentration of 5 ppm would be 8 mA, not 12 mA.

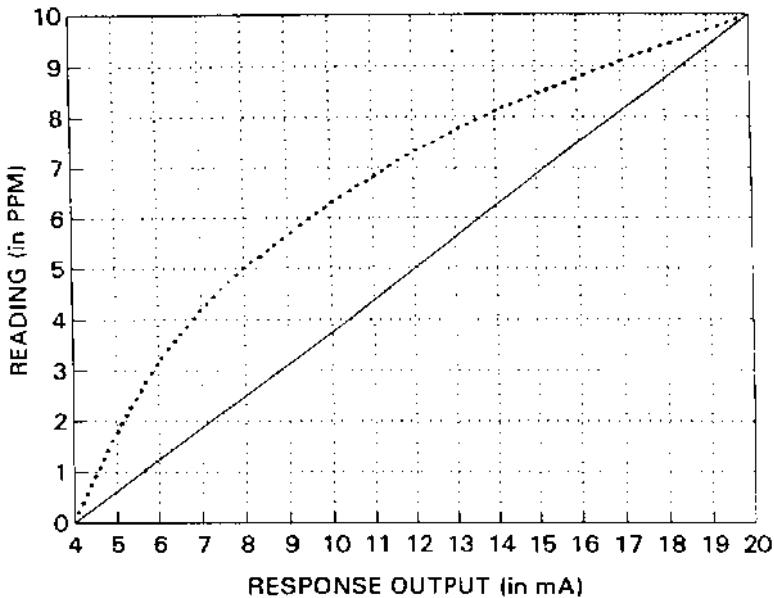


Figure 2.12. An example of a gas detector's response output. The solid line indicates a linear response output characteristic. The dotted line indicates a non-linear response output characteristic.

Interferents. A target gas concentration measurement can be distorted by interferents. An interferent is any material or condition which can affect the true measurement of the actual target gas concentration. Positive interferents are those materials and conditions which add to a target gas's concentration measurement so the gas detector's measurement reading is higher than the actual, true concentration value of the target gas. Figure 2.13 shows the effect of a positive interferent. Negative interferents mask a target gas's concentration measurement so the gas detector's measurement reading is lower than the actual concentration value of the target gas. Figure 2.14 shows the effect of a negative interferent. Some positive interferents can cause a false measurement reading by the gas detection instrument even in the total absence of the target gas. Thus, positive interferents can be the potential source of frequent false alarms in a gas detection system. Negative interferents are far more dangerous since it is possible that dangerous concentration levels of a target gas can exist before the gas detector reaches an alarm condition.

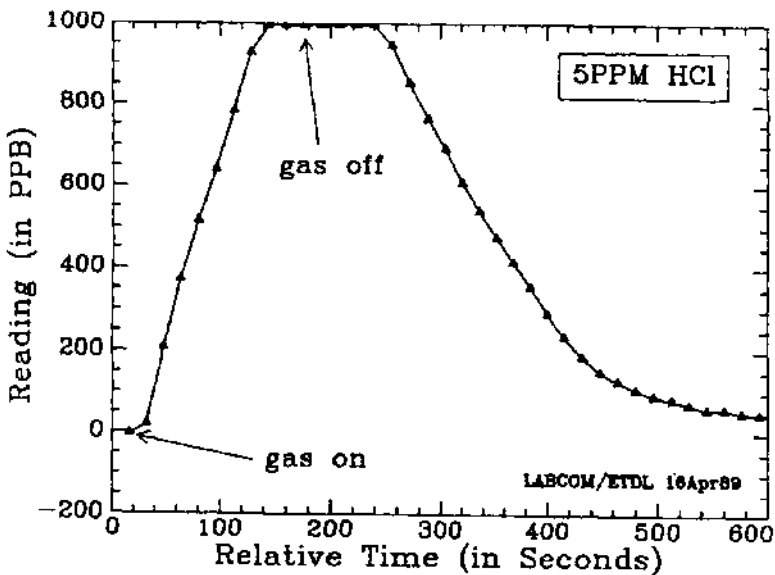


Figure 2.13. An example of a positive interferent effect on a Brand W arsine gas detector exposed to a 5 ppm environment of HCl. (This data was originally presented at the Fall 1989 NE Regional SSA Meeting held in Red Bank, NJ.)

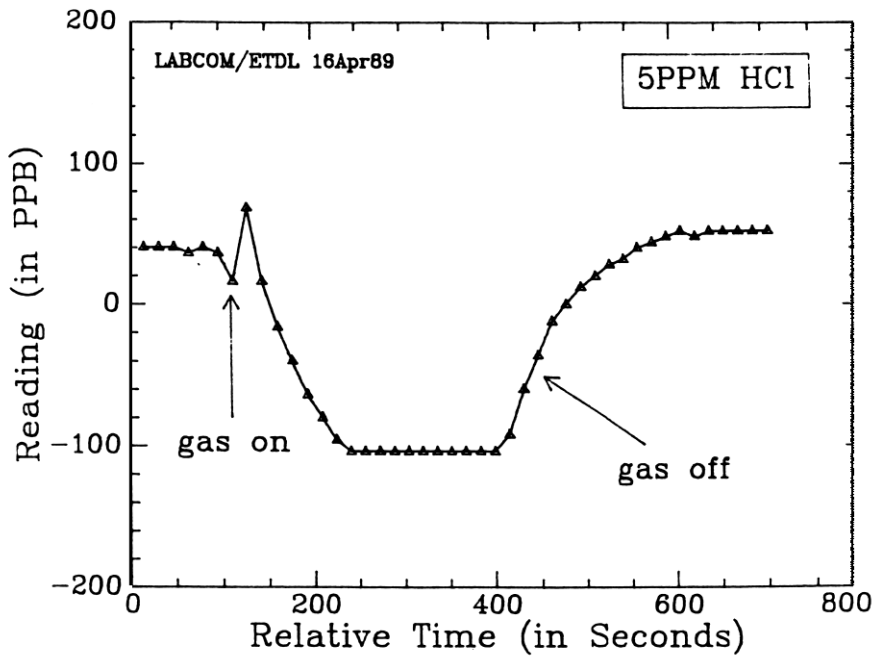


Figure 2.14. An example of a negative interferent effect on a Brand X arsine gas detector exposed to a 5 ppm environment of HCl. (This data was originally presented at the Fall 1989 NE Regional SSA Meeting held in Red Bank, NJ.)

Alarm. The alarm functional component represents the obvious output from any gas monitoring system. This ability to advise occupants of an area that a potentially toxic or flammable environment is present is essential.

Alarm/Action Setpoints. Once the target gas concentration value is determined, it must be compared with predetermined alarm/action concentration levels set by the gas detection system manufacturer and/or user. These predetermined alarm/action concentration levels correspond to the governing policies or guidelines dictated by specific environmental, health and safety codes or organizational requirements in order to protect the health and well-being of personnel and the environment. Typically, an alarm/action concentration level is based on the IDLH, STEL, PEL, or TLV values of the specific target gas. In the case of oxygen, this may also refer to a critical minimum concentration limit to protect personnel from asphyxiation in an oxygen-deficient atmosphere.^[93] Several alarm/action

setpoints are sometimes used to warn individuals of potential health and environmental hazards before they occur, or the actual increasing severity of a hazardous situation. The comparison of target gas concentration to alarm setpoint can be performed within the actual gas detection instrument or in a higher level, upstream computer system remotely located in a central control center.

Upstream Communications. There are many physical means by which this target gas concentration value can be transmitted, including optical and mechanical techniques, but the most common is through electrical communications.

Electrical signals can be either analog or digital. Digital communication signals usually require an analog-to-digital conversion process since most analyzers produce a direct analog output response to a target gas concentration. These signals may travel only a short distance (as small as an inch) if the target gas concentration comparison to alarm setpoints occurs within the same gas detection instrument. The signals might travel as long as a mile if the gas detector is located in a remote monitoring location while the target gas concentration comparison to alarm setpoints is made in an upstream personal computer system located in a far-away main control center. Some typical medium-to-long distance electrical communications protocol include 4 to 20 mA, 0 to 1 V, 0 to 10 V, and RS-232 signals.

Whatever means of transmission is used to communicate the target gas concentration, it must be reliable and protected against false signal interferences (such as electrical “noise”) when using electronic communications.

Action. The action functional component represents all the possible pre-determined response actions which should occur upon reaching a pre-determined alarm/action setpoint. Response actions could include activation of audible and/or visual alarms, annunciation of specific alarm conditions and locations, hardcopy of past and present gas concentrations or environmental conditions, notification of local and remote emergency response personnel, activation of emergency safety control systems^[94] through relays such as gas supply source and equipment shutoffs, HVAC system shutdown with 100% exhaust, or one-pass cleanroom air flow operations. The predetermined response actions should follow established emergency response procedures and policies of your organization. Also, all applicable environmental, health, and safety regulations should be followed to protect both personnel and the environment and minimize the extent of any hazardous situations.

Selection of a Toxic Gas Monitoring System. As previously stated, continued revisions and additions to national, state, and local fire and building codes along with modifications to environmental, health, and safety regulations will govern the use of toxic and hazardous materials in both the private and public sector. The requirements for and the selection of a toxic gas monitoring system will be heavily driven by all these applicable rules and regulations and their interpretation by local code enforcement officials.

Factors such as the variety and amount of hazardous materials being used, type/layout/age/location of facility, quality and number of support personnel, internal risk management and organizational policies, and available funding will also impact the design and selection of a toxic gas monitoring system. For example, the requirements for a toxic gas monitoring system from a small, non-production research and development facility's viewpoint can be quite different than a large semiconductor foundry.^[95] The following section will highlight and discuss several key issues and decisions that must be addressed before choosing a specific gas detection technology or system.

Single or Multi-Point. The specific areas (sometimes called sample points) which must be monitored must first be defined. Identification of these areas can be code driven but are generally established through a comprehensive risk management assessment of each process, operation, equipment, facility, and/or personnel. Typical areas might include gas source cabinets, gas distribution systems, process equipment, researcher or operator workspace, vacuum systems, equipment exhaust, and central exhaust discharge points. Figure 2.15 shows potential monitoring points for LPCVD and plasma-etch operations. The location of each sample line or placement of each gas detector within these areas is dependent on the physical characteristics of each area as described in Fig. 2.16. Open area leaks, enclosed area leaks, and leaks within processes must be addressed on a case by case basis. The impact of air flow, weight of the target gas relative to air, and possible dilution effects must all be considered.^[96] The types of material requiring monitoring in each of these areas, with corresponding acceptable response times, must also be determined. *Response time*, sometimes called *detection speed*, is defined as the time delay between release of a target gas in the monitored area and the detection, identification, and alarm by the toxic gas monitoring system. Selection of a response time can be influenced by the toxicity or hazard of the gas, the number of personnel who could potentially be exposed to the material, and the frequency and duration of area occupancy. If "instantaneous" response is required, single point gas detectors located in the immediate area

may be required. If a longer delay in response time can be tolerated, multi-point gas detectors might be an alternative. The choice between single point and multi-point gas detectors is not easy. While single point gas detectors usually provide fast response, they can also be very expensive to initially purchase and also maintain if a large number of them are required to adequately protect a facility. The main cost of a gas detector is in the actual sensing unit with its associated electronics. If a multi-point gas detector can cover the same area as several single point gas detectors using the same detection technology, both up-front and future maintenance cost savings usually become obvious. When comparing a multi-point gas detector with several single-point gas detectors of different technologies, the cost savings may not as clear.^[97]

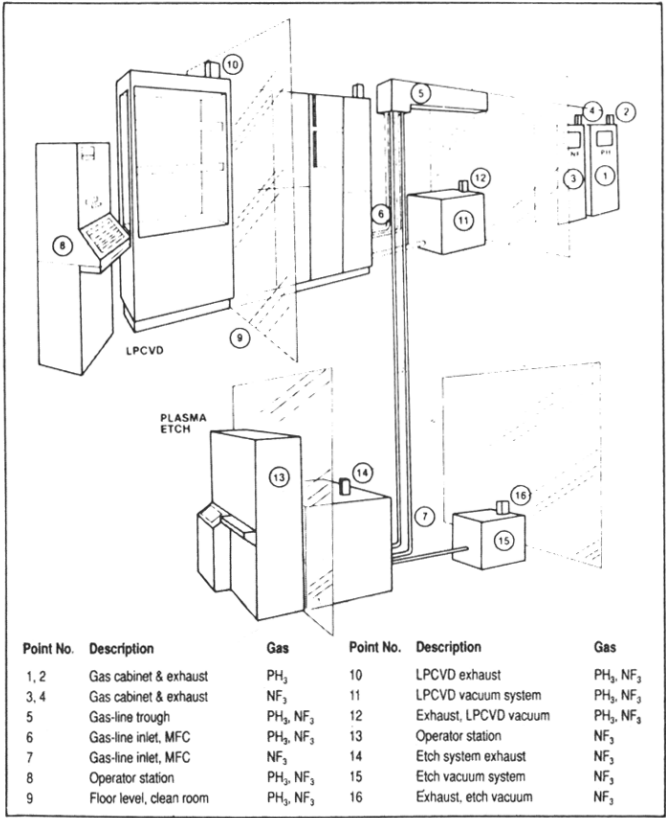


Figure 2.15. Potential monitoring points for LPCVD and plasma-etch operations using PH₃ as dopant and NH₃ as etchant and cleaning agent. Monitoring points should be placed at all potential sources of leaks. It may be necessary to monitor more than one gas at a specific location. (Reprinted with permission from the publisher, Lake Publishing Corporation.)

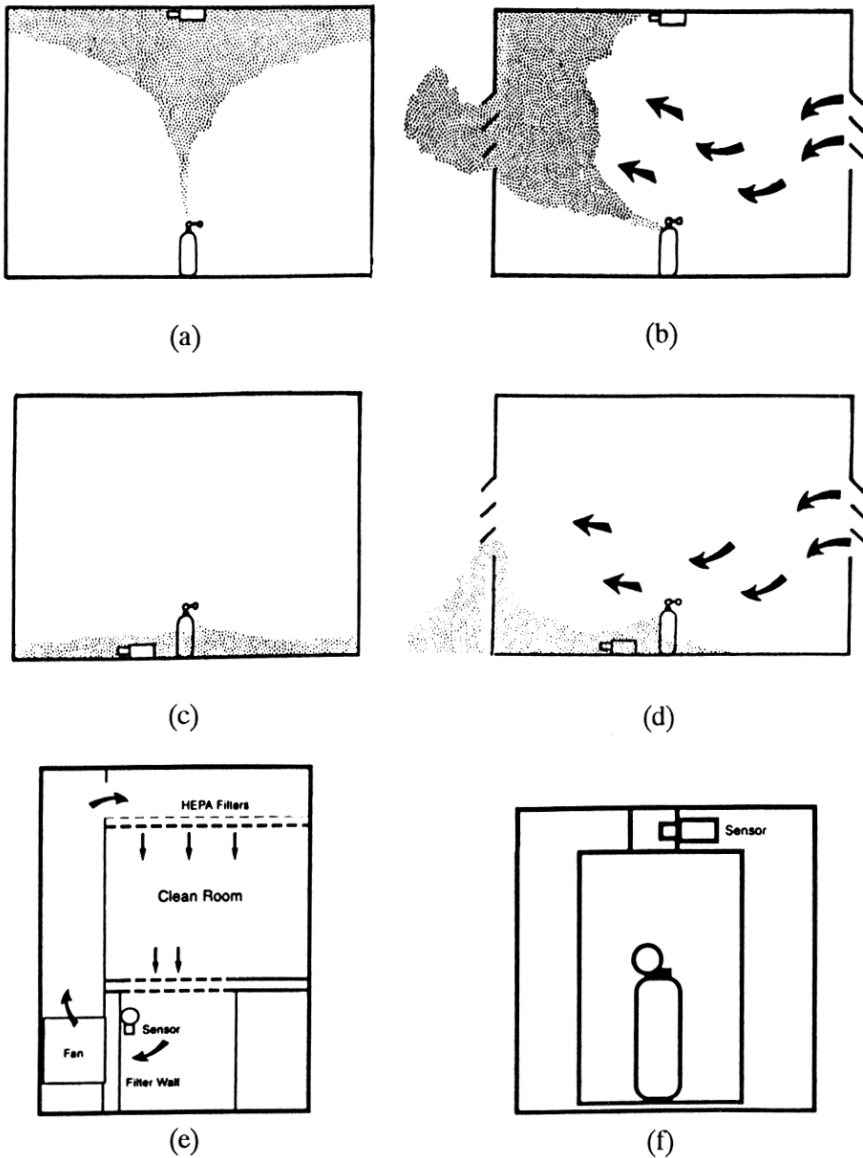


Figure 2.16. Potential locations for gas detectors based on gas vapor density and physical area characteristics. Gas density lighter than air in (a) still air, and in (b) moving air. Gas density heavier than air in (c) still air, and in (d) moving air. (e) In a typical cleanroom, the sensor should be located at the return filter wall, so it samples all air passing by during circulation. (f) In a ventilated gas cabinet, the sensor should be placed in the exhaust duct. (Figures (a)-(d) reprinted by permission, © 1981, Instrument Society of America, ISA Transactions, 20(2). Figures (e) and (f) reprinted with permission from the Semiconductor Safety Association.)

For example, a mass spectrometer is a multi-gas analyzer that can usually identify a large variety of different chemical compounds including organics, acids, and hydrides. If ten different areas must be monitored and the same three target gases must be monitored in each of these areas, there are several potential monitoring solutions. A 30-point, sample draw, multi-gas mass spectrometry gas detection system with over thirty sampling lines might meet this requirement. Or, three 10-point, sample draw, single-gas paper tape gas detection systems, each monitoring a different target gas, might also meet this requirement. Finally, thirty individual single-point, single-gas, gas detectors integrated into a central toxic gas monitoring system could also meet this requirement.

As discussed under the transport section, the length of each sample line and number of sample lines which must be scanned by a multi-point gas detector, can seriously affect response time.

For example, if the acceptable response time is 15 seconds or less and:

Number of points to be monitored - 8

Number of materials to be monitored - 1

Min. time to analyze a single sample line - 30 seconds

then:

Maximum, worst case response time is 4 minutes (which may be unacceptable)

Thus, a combination of single point and multi-point gas detectors integrated into a central toxic gas monitoring system might be required to effectively and efficiently protect a facility and its occupants.

Gas Detection Technology. The choice of gas detection technology is also a complex decision and can be influenced by the choice of single point or multi-point sampling since that choice can lock you into a certain group of commercially available gas detectors or monitoring systems.

Direct or Indirect Gas Detection. In selecting a gas detector, it is important to note whether a detector responds to the target gas directly or indirectly. For example, in monitoring for phosphorous pentafluoride, an indirect gas detector might detect hydrogen fluoride (a by-product created when phosphorous pentafluoride comes in contact with moisture in the air). In order for this indirect gas detector to provide satisfactory protection, the proper environmental conditions, such as a sufficient humidity level, must exist in the monitored area.^[98] Sometimes gas detectors are selected to monitor for the presence of a carrier gas used in conjunction with the actual

target gas. Care must be taken with this approach since carrier gas detection characteristics and limits relative to the target gas may not afford adequate protection.

Identify All Interferents. The type and affect of interfering gases or vapors can also influence the choice of gas detection technology. A gas detector, with an interferent response characteristic similar to Fig. 2.17, would probably be a poor choice in an area where equipment and other surfaces are wiped down with an isopropyl alcohol solution, and could be the source of frequent false alarms.

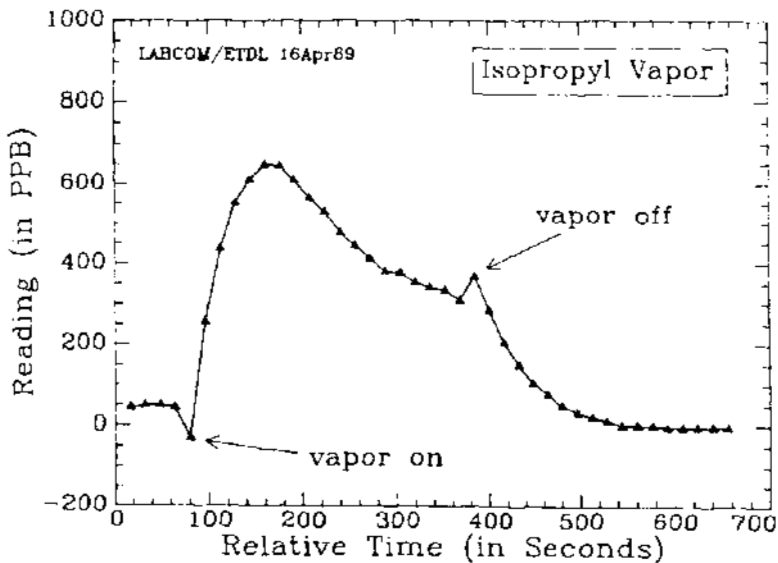


Figure 2.17. An example of an interferent effect on a Brand Y arsine gas detector exposed to isopropyl alcohol vapor. (This data was originally presented at the Fall 1989 NE Regional SSA Meeting held in Red Bank, NJ.)

Sensitivity and Resolution. The detection sensitivity and resolution of a gas detector must match the monitoring requirements for the target gas and provide an acceptable margin of safety in the event of a toxic gas leak. If a gas with a TLV of 50 parts per billion (ppb) requires monitoring and the established alarm setpoint is one half the TLV (i.e., 25 ppb), a gas detector

with sensitivity and resolution in parts per million would be totally inadequate for the application.

Factors such as stability, detection speed, recovery time, frequency and methods of calibration, required maintenance, and reliability can also influence the selection of a specific gas detection technology.

Stability and Drift. A gas detector must be stable and have a constant zero-level output signal. If this output signal varies in magnitude over time, independent of the presence of a target gas, false alarms or readings lower than the actual concentration value of a target gas can occur.

For example, a single point detector has a constant zero-level output signal of 4 mA with a linear output, a maximum scale value of 10 ppm at 20 mA, and an established alarm level of 2.5 ppm at 8 mA. In the presence of a target gas at a concentration of 2 ppm, the signal output of this detector should be 7.2 mA. If the zero-level output signal drifted higher by 1 mA, from 4 mA to 5 mA, the detector would then have an output signal of 8.2 mA in the presence of a target gas at a concentration of 2 ppm. Thus, a false alarm condition would exist due this zero-level output signal drift. Similarly, if the zero-level output signal had drifted lower by 1 mA, from 4 mA to 3 mA, the detector would have an output signal of 7.8 mA in the presence of a target gas at a concentration of 3 ppm. Thus, no alarm would have occurred, even though the target gas concentration exceeded the established alarm level of 2.5 ppm. If the detector's output was not linear, these examples could have been much worse.

Drifts can be caused by interferent gases or vapors, adverse environmental conditions, unstable conditions within the actual gas detector (such as within the electrolyte solution of an electrochemical gas detector), and/or the gas detector's associated electronics and power supply.

Environment Effects. Sometimes environmental conditions or changes can detrimentally affect the performance of both direct and indirect gas detectors. Changes or extremes in temperature and humidity, high velocity air flows across the gas detector, and/or high airborne particulate concentration can serve as sources of unwanted interference or drift as shown in Fig. 2.18. This can be a critical factor in selecting a specific gas detection technology for a specific application. For example, gas detectors which show large zero-signal level drift under changing humidity conditions would probably be a poor candidate for either outdoor or non-environmentally-controlled indoor locations in the Northeast United States where large, rapid swings in humidity during the summer months are common.

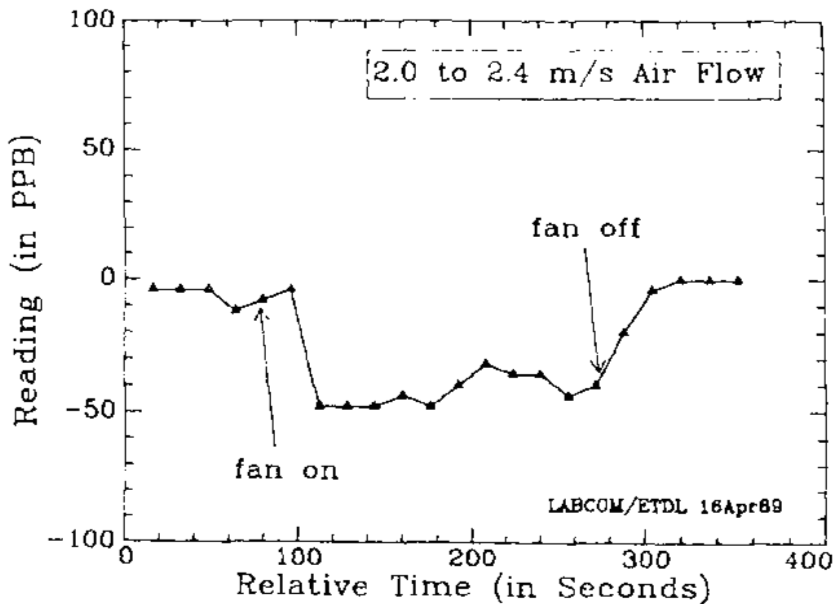


Figure 2.18. An example of sensor drift due to a change in air flow across a Brand Z arsine gas detector. (This data was originally presented at the Fall 1989 NE Regional SSA Meeting held in Red Bank, NJ.)

Detection Speed. Detection speeds can differ between various gas detection technologies and may significantly contribute to the overall response time of a toxic gas monitoring system. The selection of a specific gas detection technology can have a major impact if “instantaneous” response times are required.

Recovery Time. Certain gas detection technologies require time to recover or “clear” after exposure to a target gas or interferent. Detection accuracy can be severely affected if a second gas leak occurs in the gas detector’s monitoring area before it has recovered from the first gas leak. Thus, recovery time may influence the selection of a specific gas detection technology.^[99]

Calibration and Maintenance. Some gas detectors require involved maintenance procedures and frequent calibrations, while others require little or no maintenance and infrequent calibration. Complex maintenance and calibration procedures usually require skilled personnel

commanding higher labor charges. For example, some liquid-type electrochemical cell gas detectors require replacement of the electrolyte solution and diffusion membrane cap every month and must then be calibrated. This translates into both material and labor costs. Some paper tape gas detectors require tape replacement every few weeks. While swapping tapes is usually a short and easy job, the long term material costs associated with purchasing replacement tapes should be a consideration. Some detectors require stabilization time after maintenance before calibration can be performed. Since maintenance and calibration procedures are usually performed concurrently by the same individual, increased labor costs can be incurred due to this additional waiting time. Additionally, built-in internal diagnostics can aid maintenance personnel in quickly determining and correcting gas detector problems, reducing service time and associated labor costs. Long term calibration and maintenance costs are directly dependent on the technology and number of gas detectors involved, and therefore can influence the choice between single-point, multi-point, or combination of single point and multi-point gas detectors used to monitor a facility.

Direct and Indirect Calibration. A number of different methods can be used to calibrate a gas detector. The direct calibration method exposes the gas detector to known concentrations of the actual target gas and calibrations are made against these known concentrations. Indirect calibration methods use calibration techniques, other than exposure to the actual target gas, that can be correlated to specific concentrations of the target gas. For example, some paper tape gas detectors use a verification card placed in front of the detector's electro-optical system which simulates the corresponding tape stain for a specific target gas concentration. Other gas detectors might use a less hazardous positive interferent gas whose detector response directly correlates to the response for a specific target gas. Whatever calibration method is used, it must be accurate, repeatable, and reflect the detector's response to the actual target gas across the entire range of the gas detector.

Contractor Support. Availability of contractor support services for both gas detectors and upstream system hardware and software must be assessed. Service advice via a toll-free 800 phone number, operation and maintenance training of user personnel, and quick response on-site field service assistance are all desirable support features.

Thus, the choice of gas detection technology, or combination of technologies, is dependent on many variables and factors.

Pre-Planning. Assessing potential environmental conditions and interferent gases or vapors that can exist in the area to be monitored, defining gas detection sensitivity and resolution requirements, and establishing maintenance and calibration procedures, frequency, and costs requires in-depth, thorough information-gathering and pre-planning. While gas detector manufacturers can supply a large amount of background information on topics like interferent materials, accuracy, sensitivity, detection speed, recovery time, calibration, and maintenance, they may be somewhat reluctant in citing areas of poor performance. It is strongly suggested that the established user base be queried and that actual hands-on testing of potential gas detector candidates be performed prior to final selection and purchase of any gas detector or toxic gas monitoring system.

Gas Detector Integration. Once the areas to be monitored are defined, the type of gas detection technologies and number of gas detectors are selected, location of each specific sample line or gas detector in each area are determined, and alarm/action setpoints are chosen, the gas detectors must be incorporated into an integrated continuous toxic gas monitoring system. This toxic gas monitoring system must be created within the framework established by regional or local codes, local code officials, internal risk management practices, organizational policies, processes and operations, equipment and facilities, and qualifications and availability of personnel. The system might be centralized, distributed, or a combination of centralized and distributed. Where single point gas detectors are used, gas concentration values can be sent to a higher-level upstream computer system located in a central control center where comparisons of target gas concentration to alarm setpoints are performed. As previously described under the alarm functional component, the target gas concentration value is usually transmitted by electrical or fiber optic communications. Communications lines must be installed between the single point gas detectors and upstream system. In order to avoid running a large number of cables, several gas detector output signals within a local area can be multiplexed through a single communications line. Some single-point gas detectors, and most multi-point gas detectors, perform the comparison of target gas concentration to alarm setpoint locally and then transmit their results to the upstream system.

Alarms. As discussed under the action functional component, once the target gas concentration exceeds the pre-determined alarm/action setpoint, response actions are initiated. Audible and/or visual alarms must be installed where appropriate to warn all affected personnel of an alarm

condition. Different warning signals can be used to distinguish between low and high level alarms, possibly requiring evacuation of a local area or an entire building. While many single point and multi-point gas detectors have both a visual gas concentration reading and combination alarm light and buzzer/horn, they usually do not provide sufficient warning outside the immediate gas detector area. Therefore, gas detector alarms are, in many cases, supplemented by more visible and audible alarms activated locally by an alarm relay connected to the detector or a signal sent from the upstream system. Communication lines, switches, relays, etc. must be installed from the action-determining systems to the appropriate alarm locations. In addition to warning personnel in the affected areas, emergency response personnel must also be notified. Emergency response personnel can be contacted by on-duty personnel manning the central facility control center once they have received notification from the toxic gas monitoring system that an alarm condition exists. Emergency response personnel can also be notified through a pager/call-back system that is automatically initiated by the toxic gas monitoring system. When emergency response personnel are paged, they can interactively query the toxic gas monitoring system via a touch-tone phone and computer voice status messages generated in real time.^[100]

Record Keeping. A record of past and real-time present gas detector measurements during an alarm condition may be required. Target gas concentration values can be saved on hardcopy or magnetic media. For example, a small thermal printer contained within a single point or multi-point gas detector can record the time, date, and concentration values on a routine periodic basis or frequently during an alarm condition. Similarly, an upstream system could use a line printer or computer data file (saved to disk) to record the same information. Since this information is very useful to emergency response personnel, hardcopy or on-line computer query capability of disk-saved data (both locally near the alarm site and at a remote, central control center) is desirable.

Emergency Safety Systems. Activation of emergency safety control systems through alarm relays controlled by the gas detector or upstream system must also be initiated when an alarm condition exists. Usually, gas flows from the gas supply source and to each piece of equipment using the detected target gas are shutoff by automatic gas valves directed by alarm action communication signals. It is generally desirable for the area affected by a gas leak to remain under negative air pressure (with respect to adjacent, non-contaminated areas) and may help to prevent the leak from spreading. Sometimes, this condition can be

achieved by shutting down the normal HVAC supply air system and either leaving on the exhaust to normally exhausted equipment and/or turning on a secondary exhaust system to remove contaminated air from the affected areas. To minimize contamination entering controlled-particle environments, some facilities have been designed for single-pass cleanroom air flow operations during a gas alarm condition. Communication links and control systems must be established between the gas detector and/or upstream system and emergency safety systems. In certain geographic regions, seismic detection or earthquake safety systems can also activate emergency safety systems utilizing the same communication links and control systems.^[101] Both the toxic gas monitoring system, including all gas detectors and associated electronics, and emergency safety and control systems must not be affected by utility failures. Provisions must be made to provide secondary utility support such as battery backup or connection to an uninterruptable power source.

Data Management System. During the late 1980s and early 1990s, tremendous improvements in the use of multiplex communication techniques and personal computer (PC) based data management systems for toxic gas monitoring have greatly reduced the integration burden of toxic gas monitoring system designers, installers, and users. Standard communication protocol, off-the-shelf communication cards easily installed in PCs, and application software based on common database and graphics software packages have provided a useful, user-friendly, cost-effective means of monitoring an entire facility from a single control station. As shown in Fig. 2.19, displays of facilities and equipment layout via high resolution computer graphics indicating alarm locations, real time notification of alarm/action events via on-screen pop-up windows, and close-up, detailed illustrations of system components and functional schematics via mouse-driven zoom graphics all represent desirable system features. Data acquisition of alarm conditions (noting date, time, locations, concentrations, and alarm/action setpoints), zero-level output signal drifts, eight-hour time weighted average concentrations, calibration and maintenance history, and planned or unplanned system interruptions allows the user to document all important monitoring events and, where necessary, accumulate this information for later analysis.^[102] Alarm history, gas detector reliability, interferent trends, and long-term system performance are just a few examples of the reports that can be generated and analyzed using many of these PC-based data management systems. Many facilities also have automated computer controlled building management systems. Many toxic gas monitoring data

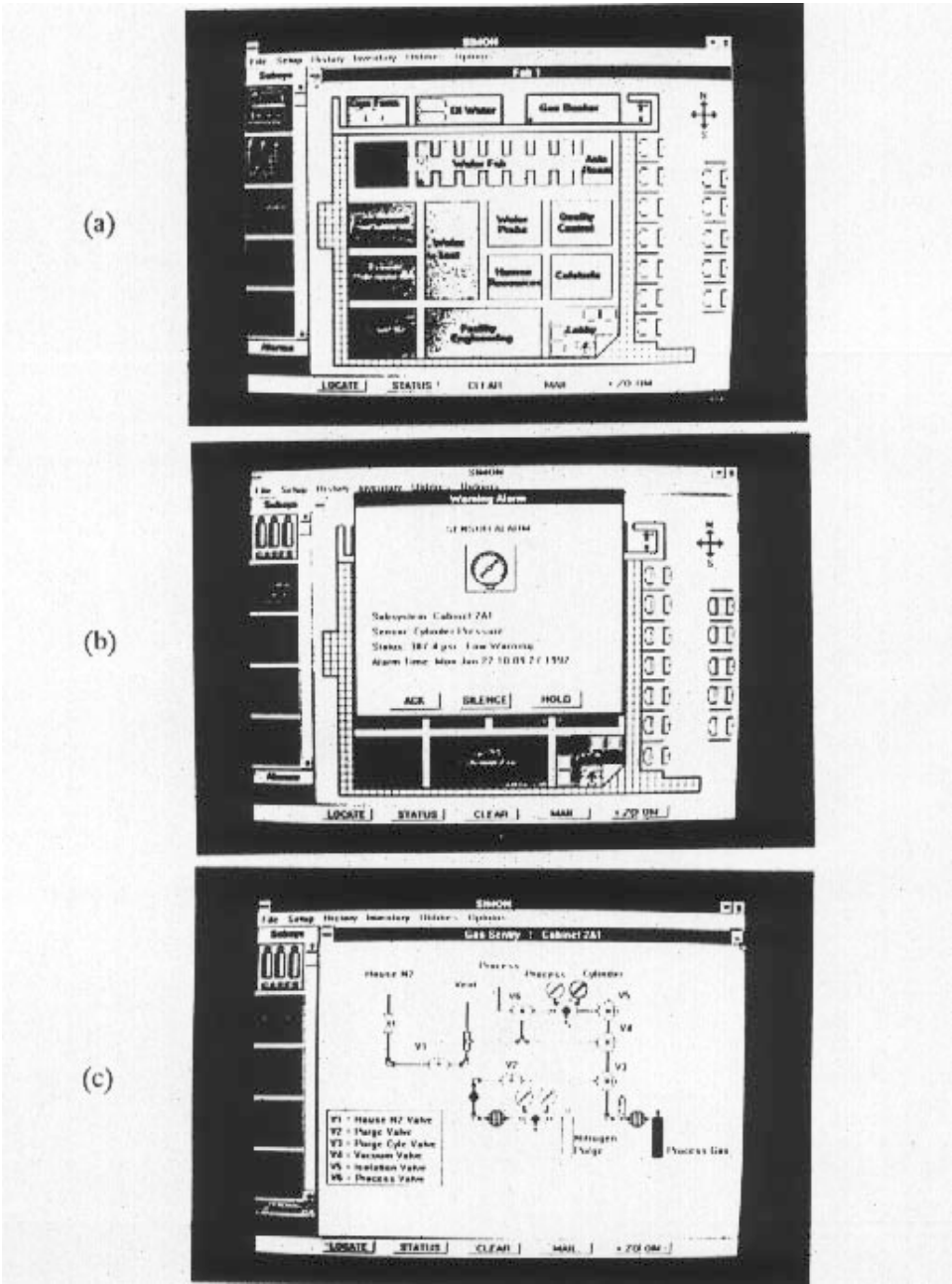


Figure 2.19. An example of high resolution computer graphics used by a toxic gas monitoring data management system to show (a) a primary facility map, (b) “alarm” window, and (c) zoom into gas cabinet schematic for sensor in “alarm.” (Reprinted with permission from SPAN instruments.)

management systems can be further integrated with these building systems to provide a single, comprehensive facility management/control system. When selecting a toxic gas monitoring system, evaluation of these capabilities should be considered. The size of the established user base, whether the software is custom generated or available off-the-shelf, and the level of difficulty and cost of upgrade, expansion or modification in the future as system requirements change are some of the factors which can affect this decision.

Future Upgrades. Just as the upstream data management system must be easily and cost-effectively upgraded or modified, the entire toxic gas monitoring system should be planned with future change and/or expansion as an underlying design requirement. In facilities where technological focus and types of materials used are constantly changing, flexibility in rearranging gas detection locations and substitution of gas detectors with different technologies are essential to quickly and economically meeting the toxic gas monitoring needs of an organization. Forethought to layout and accessibility of sampling and communication lines, building in sufficient capacity for expansion to accommodate the monitoring requirements of new gases and/or additional monitoring locations, and change/expansion of associated alarm/action systems must be addressed early in the design cycle of any toxic gas monitoring system.

Role of Portable Gas Detectors. Most toxic gas monitoring systems integrate permanently installed gas detectors in fixed locations throughout a facility. Portable gas detectors, based on several of the technologies found in the analysis functional component section, provide an important complementary function to these permanently installed gas detectors. Portable gas detectors provide temporary continuous monitoring protection during equipment and toxic gas monitoring system maintenance, are used to check for leaking gas cylinders upon delivery, and assist emergency response personnel in determining actual gas leak locations. Most of the technology and performance information related to permanently installed gas detectors previously discussed is also relevant to portable gas detectors.

Future Trends. Gas detection techniques, monitoring systems, data management systems, emergency safety systems, and level of integration will continue to improve through both technological advancements and product enhancements. Gas analyzer response characteristics, including sensitivity, resolution, stability, detection speed, and recovery time, will be refined and improved. Additions or modifications to existing gas detectors

will be made to reduce or eliminate the effects of known interferences and environmental conditions or changes. Developments of existing gas detection techniques will increase reliability and reduce the frequency and/or level of required maintenance. New gas detection techniques will be developed and introduced into the marketplace. For example, current state-of-the-art quartz resonators are being used to develop high frequency, high stability clock/time sources with accuracies approaching one part per trillion. These quartz resonators are fabricated in ultra-clean environments and are extremely sensitive to surface impurities. A single monolayer of contamination can easily shift the output frequency of a resonator by one in a million. It is conceivable that a gas detector based on this contamination-related frequency shift effect could be developed with both excellent sensitivity and reliability.

Emerging electronic device technologies, such as micro-electromechanical, high energy/micro-magnetic or biosensor devices, coupled with advances in electro-optics, may provide the foundation for completely new gas detection techniques or order-of-magnitude smaller, next-generation gas detectors such as a mass-spectrometer-on-a-chip. These future gas detection techniques, along with the development of distributed intelligent control systems, may enable the use of an extremely large number of highly sensitive, multi-gas, single-point gas detectors to economically monitor all potential locations of interest. The development and improvement of electro-optical detection technologies, such as passive IR and laser techniques, may offer promising near-term and longer-term candidates for stand-off, remote chemical detection and identification required by applications such as perimeter monitoring of fence-line emissions, in situ exhaust monitoring of stack effluents or fugitive process emissions,^[103] and hazardous waste disposal site surveillance.

As computer information systems evolve, data management techniques and systems for toxic gas monitoring will also improve. Faster, user-friendly, icon or voice-driven software utilizing high resolution graphics on future large area or next generation three-dimensional displays will provide both the user and emergency response personnel with useful, real-time leak status information. Standardized optical-bus or wireless communication techniques will allow users to rapidly and cost effectively modify or expand the number and location of single point gas detectors used in their toxic gas monitoring system. Comprehensive integration of the toxic gas monitoring system with all other equipment and building control and monitoring systems, along with voice, video and data communications, will permit true "Smart Facilities."

One fact is for certain: both gas detectors and toxic gas monitoring systems must continue to improve to keep pace with future revisions and additions to national, state, and local fire and building codes and modifications to environment, health, and safety regulations for effective and affordable monitoring and protection of personnel, environment, and facilities.

4.3 Surface Contamination

The subject of wipe sampling is not well defined in standard industrial hygiene reference books. The use of wipe sample data is open to interpretation,^[104] however, wipe sampling is of interest to the semiconductor industry because of certain metals used in the manufacturing process such as arsenic, antimony, chromium, and lead. A variation on this technique is also used in the evaluation of surface contamination with sealed radioactive sources (see Ch. 3.0: Physical Agents, Ionizing Radiation, Radioactive Materials).^[105]

Ordinarily wipe samples are used to semi-quantitatively measure surface contamination of materials that may present an ingestion hazard. At least in theory, wipe samples can also be used to establish the potential for skin absorption and/or hazardous skin effects, but this is normally very secondary to their use as indicators of ingestion of toxic metals.

When wipe sample data show “high” levels of contamination, the surfaces should be cleaned more frequently or other steps should be taken to reduce the contamination. What is considered “high” is the debatable issue. As an example, the U.S. Federal OSHA lead standard^[106] states: “All surfaces shall be maintained as free as practicable of accumulations of lead.” Similar wording is also contained in the OSHA arsenic standard.^[16] The following approach to wipe sampling is being used in the semiconductor industry:

- Take wipe samples of work surface to establish the presence of the suspected contaminant
- Thoroughly clean the surface and/or change work procedures to prevent contamination
- Resample to determine post-clean contamination levels and to establish a comparison for future sample results

One simple, generic approach to establishing an acceptable surface residue level is:

- Take the airborne standard (e.g., 10 $\mu\text{g}/\text{m}^3$ for arsenic, 50 $\mu\text{g}/\text{m}^3$ for lead...)
- Use the average amount of air inhaled during an 8-hr shift (i.e., 10 m^3)
- Assume the total retention of the inhaled material during the shift and calculate an effective dose (i.e., 100 μg for arsenic, 500 μg for lead...)
- Apply this calculated value to the typical surface area that has been wiped in obtaining the sample (e.g., 100 cm^2), to obtain the “standard” for the material (e.g., 100 $\mu\text{g}/100 \text{ cm}^2$ for arsenic, 500 $\mu\text{g}/100 \text{ cm}^2$ for lead...)

This procedure is also the basis of a California’s OSHA wipe sample requirement in the 4,4'-Methylenebis(2-Chloroaniline), MBOCA, standard. Section 5215(c)(2)(C) of California’s *General Industry Safety Orders* states: “Accessible surfaces which employees are likely to contact shall not exceed 100 μg MBOCA per 100 cm^2 of surface. For regulatory purposes, the average of at least five wipe samples shall be used; if one sample exceeds the mean by a factor of 10 or more, it will be rejected as not being representative.” MBOCA has a CAL-OSHA Permissible Exposure Limit of 10 $\mu\text{g}/\text{m}^3$.

4.4 Biological Hazards

There are no biological hazards uniquely associated with semiconductor manufacturing. Those that are present in the industry are generic to typical office and manufacturing environments. These include microbiological agents such as bacteria, molds, and fungi caused by moisture incursion from sources such as stagnant water in HVAC air distribution systems and cooling towers. Other potential—but less likely—agents include mites, pollen, algae, insects, animal dander, and bird droppings.

Sampling for micro-organisms is not recommended in most instances because there are no generally accepted guidelines for what concentrations or amounts of bioaerosols are appropriate. One exception to this rule of thumb is when there has been a medical diagnosis of Legionnaires’ disease or Pontiac fever. Sampling for bioaerosols is a highly specialized area of IH monitoring. Persons with particular expertise in this area should be consulted.

Elimination of standing water and effective water treatment are the best controls for preventing microbiological contamination. All ventilation systems should be maintained to minimize or eliminate standing or stagnant water.

If the relative humidity in occupied spaces is greater than 60%, the growth of allergenic or pathogenic organisms will be stimulated. If the humidity in occupied spaces and low velocity ducts and plenums exceeds 70%, fungal contamination (e.g., mold and mildew) can occur. Humidity levels below 20% are associated with increased discomfort and drying of the mucous membranes.^[105]

In modern semiconductor fabs, overall relative humidity is controlled within a specific range (e.g., 35–55% RH). On the low side, humidity is controlled to minimize static electricity. On the high end, humidity is controlled primarily to maintain consistency in photoresist sensitivity.

Air handling unit condensate pans should be designed for self-drainage to preclude the buildup of microbial slime. Also, provision should be made for periodic in situ cleaning of cooling coils and condensate pans with an anti-microbial agent. Air handling and fan coil units should be easily accessible for inspection and preventive maintenance.

Steam is preferred as a moisture source for humidifiers, but care should be exercised to avoid contamination from boiling water or steam supply additives. Cold water humidifiers are discouraged. However, if cold water humidifiers are used, the water should originate from a potable source, and, if recirculated, the system will require frequent maintenance and blow-down.

If standing water is used in conjunction with water spray in HVAC air distribution systems, it should be treated to avoid microbial buildup. Also, special care should be taken to avoid entrainment of moisture drift from cooling towers into the makeup air and building vents.

Microbiological Contamination

General Information. Five percent of NIOSH's indoor air quality (IAQ) investigations have involved some type of microbiological contamination.^[108] Even though this is not a commonly reported cause of problems in the semiconductor industry, it can result in a potentially severe health condition known as hypersensitivity pneumonitis. This respiratory problem can be caused by bacteria, fungi, protozoa, and microbial products that may originate from ventilation system components. A similar condition known as humidifier fever, most commonly reported in Europe, is also the result of

microbiological contamination in ventilation systems. Microbiological contamination has commonly resulted from water damage to carpets or furnishings, or standing water in ventilation system components.

Although a variety of disorders (hypersensitivity pneumonitis, humidifier fever, allergic rhinitis, conjunctivitis) can result from microbiological exposure, NIOSH generally has not documented the existence of these disorders on the basis of medical or epidemiological data. However, even if visible microbial growth can not be directly related to the health complaints reported, it is a problem that needs to be addressed and corrected. Also, certain recent studies appear to contradict the NIOSH experience, and indicate microbiological contamination and the resulting adverse health effects may be more prevalent, but under-reported.^[109]

Water Treatment. Bromides, chlorides (such as formulations of the alkyl dialkylbenzylammonia chloride salts), or other microbiocides can be used to control slime, algae, plankton, bacteria, and other micro-organisms in cooling towers. Cooling tower water should be regularly tested to ensure the proper amounts of biocides and corrosion inhibitor are maintained.

As biocides, bromides are four times more effective than chlorides. Also, bromides dissipate in the blowdown in half the time of chlorine compounds (since they have a shorter half-life). Sunlight and ultraviolet light break down chlorine into other compounds, and chlorine residuals are difficult to maintain due to the variation in organic level within the condenser water. Cooling towers should be cleaned out at least once a year and more often where they are located next to farming, construction, or other types of activity that increase the suspended solids in the tower water.

Mold and Mildew Problem Identification. Mold and mildew are commonly found on the exterior wall surfaces of corner rooms in regions where interior heating is used. An exposed corner room is likely to be significantly colder than adjoining rooms, so that it has a higher relative humidity (RH) than other rooms at the same water vapor pressure. If mold and mildew growth are found in a corner room, then RHs next to the room surfaces are above 70%. However, is the RH above 70% at the surfaces because the room is too cold or because there is too much moisture present?

The amount of moisture in the room can be estimated by measuring both temperature and RH at the same location and at the same time. Suppose there are two cases. In the first case, assume that the RH is 30% and the temperature is 70°F (21°C) in the middle of the room. The low RH at that temperature indicates that the water vapor pressure (or absolute

humidity) is low. The high surface RH is probably due to room surfaces that are “too cold.” Temperature is the dominating factor, and control strategies should involve increasing the temperature at cold room surfaces.

In the second case, assume that the RH is 50% and the temperature is 70°F (21°C) in the middle of the room. The higher RH at that temperature indicates that the water vapor pressure is high and there is a relatively large amount of moisture in the air. The high surface RH is probably due to air that is “too moist.” Humidity is the dominating factor, and control strategies should involve decreasing the moisture content of the indoor air.

Reference 110, “Guidelines for the Assessment of Bioaerosols in the Indoor Environment,” contains additional information on assessing and controlling microbiological contamination. Reference 111, “Building Air Quality—A Guide for Building Owners and Facility Managers,” includes additional information on molds and mildews.

Legionella. The information in this section is primarily taken from Ref. 112, “The Prevention or Control of Legionellosis—Approved Code Of Practice.” *Legionella pneumophila* is the cause of the most serious IAQ problems. It is a bacteria commonly found in nature. Optimum water temperature for growth of this bacteria is 80–90°F (27–32°C). *Legionella* bacteria can cause two distinct clinical syndromes: Legionnaires’ disease and Pontiac fever. Legionellosis is the term used for infections caused by *Legionella pneumophila* and other bacteria from the family Legionellaceae. Legionnaires’ disease is a bacterial pneumonia. It principally affects those who are susceptible due to age, illness, immunosuppression, smoking, etc., and sometimes results in death. It has an incubation period of five or six days, and only a small percentage of exposed persons contract symptomatic disease. In addition to pneumonia, the infection may involve the gastrointestinal tract, kidneys, and the central nervous system.

Legionella is also associated with Pontiac fever which is characterized by fever, chills, headache, and muscular pain. Pontiac fever has an incubation period of about 1.5 days, an attack rate of almost 100%, but is much less severe than Legionnaires’ disease.

Infection is attributed to inhaling Legionellae, either in those water droplets which are small enough to penetrate deeply into the lung, or in droplet nuclei (the particles left after the water has evaporated). Legionellae are widespread in natural sources of water. They may enter man-made systems or water services, where they can multiply under certain conditions, and if there is a means of creating and transmitting water droplets,

people in the vicinity may be at risk. Most cases and outbreaks of legionellosis have been identified in foreign outbreaks including a humidification system, industrial coolants and respiratory therapy equipment.

Experience shows that the following present a risk of legionellosis:

- Water systems incorporating a cooling tower
- Water systems incorporating an evaporative condenser
- Hot water services, except where the volume of hot water in the system does not exceed 300 liters
- Hot and cold water services irrespective of size in premises where occupants are particularly susceptible, e.g., health care
- Humidifiers and air washers which create a spray of water droplets, and in which the water temperature is likely to exceed 68°F (20°C)
- Spa baths and pools in which warm water is deliberately agitated and recirculated

Other plants and systems containing water which is likely to exceed 68°F (20°C) and which may release a spray or aerosol (i.e., a cloud of droplets and/or droplet nuclei) during operation or when being maintained may also present a risk.

Prevention is the best control for *Legionella* (e.g., no standing water and water treatment). Treatment against *Legionella* is usually effective by dosing with chemicals such as calcium hypochlorite, quaternary ammonium compounds, and dibromonitripropionamide. Hyperchlorination is only effective if residual quantities of free chlorine (greater than 2 mg/L) are maintained.

Unless there has been a diagnosis of Legionnaires' disease or Pontiac fever, sampling for *Legionella* is not necessary because there are no generally accepted guidelines for what concentrations or amounts of the bacteria are inappropriate. However, some investigators suggest:

- <1 colony forming units (CFUs) per ml of water is an acceptable concentration of *Legionella* in reservoirs
- >100 CFUs per ml requires cleaning
- >1000 CFUs per ml requires "emergency" cleaning

Identification and Assessment of Risk. If the presence of *Legionella* is suspected, a suitable and sufficient assessment should be carried out by a qualified consultant to identify and assess the risk of legionellosis from work activities and water sources on the premises and any

necessary precautionary measures. The purpose of the assessment is to enable a valid decision to be made about:

- The risk to health, i.e., whether the potential for harm to health from exposure is reasonably foreseeable unless adequate precautionary measures are taken
- What measures for prevention, or adequate control to minimize the risk from exposure to *Legionella*, should be taken

The assessment should include identification and evaluation of potential sources of risk and:

- The particular means by which exposure to *Legionella* is to be prevented
- If prevention is not reasonably practicable, the particular means by which the risk from exposure to *Legionella* is to be minimized

Where the assessment demonstrates that there is little or no risk, no further assessment or measures are necessary. However, should the situation change, the assessment should be reviewed and any necessary changes implemented.

The assessment should be reviewed whenever there is reason to believe that the original assessment may no longer be valid. For example, this may be because of:

- Changes to the plant or water system or its use
- Changes to the use of the building in which it is installed
- The availability of new information about risks or control measures
- The results of inspections indicating that control measures are no longer effective

Preventing or Minimizing the Risk from Exposure to Legionella.

Where the assessment shows that there is a reasonable risk, the use of water systems, or operations that lead to exposure should be avoided. Where this is not feasible, there should be a written plan for minimizing the risk of exposure. The plan should be sufficiently specific and detailed. In particular, it should contain information about the operation that is necessary to minimize the risk from exposure.

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